

Software-as-a-Graph: When Does Graph Learning Improve Cascade-Impact Ranking in Publish–Subscribe Systems?

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Abstract

Publish–subscribe middleware decouples distributed components, obscuring failure propagation before deployment when telemetry is unavailable. Graph learning has been applied to component criticality, but whether accuracy stems from the learning algorithm or the dependency representation remains unexamined. We present Software-as-a-Graph (SaG), which derives explicit dependency graphs from declarative architecture specifications. We evaluate analytical, hybrid, and learned rankers across twelve synthetic architectures, three failure simulators, and five stylized open-source system models. An order- k reference criterion separates rankings that restate a simulator’s propagation rule from those predicting it, preventing circular validation.

On a reachability simulator, graph attention on the derived dependency graph reaches Spearman’s $\rho = 0.748$, +0.072 above the same model with reverse edges on the raw multigraph, where co-primary contrasts were null. Counting direct dependents ($\rho = 0.764$) is not significantly different from the learned model. For a queue-flow simulator requiring 12.7 CPU-hours of labeling, a rate-weighted first-order approximation reaches $\rho = 0.830$ in milliseconds, exceeding a learned approximation trained on its labels ($\rho = 0.799$; exploratory); learners that start from the formula do not improve on it. In zero-shot transfer, learned models and dependency counts separate inert from active components ($\rho \approx 0.81$ vs. 0.53), though active-stratum discrimination remains weak.

On these simulators, whose propagation is first-order by construction, representation provided the predictive signal. Learning did not outperform aligned analytical references. We recommend evaluating graph learning against low-order analytical references aligned with the simulation oracle, while training-free dependency analysis enables rapid pre-deployment review.

Keywords: Dependency graphs, cascading failures, publish–subscribe, graph neural networks, reliability, software architecture, afferent coupling

1. Introduction

1.1. Motivation and Problem

Large distributed systems increasingly use asynchronous publish-subscribe (pub-sub) middleware: ROS 2 for autonomous driving [1], Apache Kafka for enterprise event streams [2], DDS in cyber-physical systems [3], MQTT in IoT [4], and cloud-native microservices [5, 6]. Pub-sub separates producers and consumers in space, time and synchronization [7]; components communicate via topics and brokers, and Quality-of-Service (QoS) policies set at deployment control reliability, durability, priority and deadlines. This decoupling also hides how failures spread. Outages and backpressure propagate through brokers, shared topics, colocated hosts and shared libraries, in sequential cascades (a failed publisher starves its subscribers, or a slow subscriber fills a broker queue) and in simultaneous blasts (a library crash takes down every service that uses it), much as overload cascades through complex and interdependent networks [8, 9]. Neither kind is visible in architecture

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diagrams or call graphs. Fault prevention and architecture evaluation belong to design time [10, 11], before deployment, when no telemetry exists, so architects must identify, from configuration manifests alone, the application services whose failures have the greatest cascading impact.

Current practice leaves a gap between service-level analysis and system dependability. Architectural properties arise from how components interact rather than from any single component [12], and architectural smells are invisible at the level of individual modules [13]: a system can have no defect in any service yet fail through hidden single points of failure or mismatched QoS contracts. The Architecture Tradeoff Analysis Method (ATAM) relies on manual input [14], static code analysis inspects services in isolation [15, 16], and chaos engineering [17], fault injection [18] and dependency tracing [19] require a running system and considerable computation [20, 21, 22]. Graph representations have been used for reliability and defect analysis [23, 24], and graph learning has recently been applied to network criticality [25, 26] and microservices [27]. It remains unclear whether such models’ performance stems from the learning algorithm or from the dependency representation it reads, and whether and when graph learning adds measurable value beyond analytical methods based on the same dependencies. Software-as-a-Graph (SaG) addresses this question by deriving explicit dependency graphs from declarative architecture specifications and enabling a controlled comparison of analytical, hybrid, and learned approaches to cascade-impact ranking before deployment.

1.2. The Software-as-a-Graph (SaG) Approach

SaG models an event-driven architecture as a typed, directed multigraph over Applications, Brokers, Topics, Execution Hosts and Shared Libraries (Figure 1), derives an explicit `DEPENDS_ON` dependency graph through publish–subscribe rules (Section 3.3), and supports analytical, hybrid and learned cascade-impact rankings on it (Sections 4 and 5.2). Labels come from three simulators on the raw topology that encode different failure notions: structural reachability (I^*), discrete-event queue saturation (I_{dyn}) and multi-criteria fragmentation (I_{comp}) (Section 4.3). Because simulators and rankers read the same architecture, some analytical rankings restate a simulator’s rule; a reference criterion (Section 4.4) treats these as analytical truncations of the simulation rule and reports them as references rather than predictors, so circular agreement is not mistaken for predictive skill. The reachability simulator is fast, serving to calibrate the method; the queue-flow simulator models message rates and queues, providing an evaluation bed for analytical versus learned surrogate modeling.

1.3. Findings in Brief

Explicit dependency representations provide most of the predictive signal for the reachability and queue-flow simulators.

Making dependencies explicit enables graph learning. On the raw multigraph every relation points away from Applications, so message passing never reaches the scored nodes, and the registered co-primary contrasts (a heterogeneous graph transformer, with and without the QoS channel, against the training-free baseline on that graph) were null (QoS-weighted: $\Delta\rho = +0.069$, $p = 0.266$). On the dependency graph, a graph attention network (GAT-P-QoS) reaches Spearman $\rho = 0.748$ against the reachability simulator, against 0.635 on the raw multigraph. Neither edge direction (+0.072 over a GAT that passes every raw edge both ways, Holm $p = 0.014$) nor the input features that compute part of the simulator account for the gain: with all of them removed, the dependency-graph GAT beats an equally stripped reverse-edge control by +0.231 (Holm $p = 0.0068$).

The same representation yields analytical rankings that no learned model exceeded. Counting direct dependents, publish–subscribe afferent coupling, reaches 0.764, not significantly different from the best learned model (equivalence not established). On the queue-flow simulator (12.7 CPU-hours to label the corpus), a rate-weighted first-order approximation reaches 0.830 in milliseconds; a gradient-boosted approximation trained on the simulator’s labels reaches 0.799 (exploratory), and learners that start from the formula do not improve on it: as an extra input it leaves a gradient-boosted model at its value, and as a prior it leaves a graph attention network below it (−0.018).

Learning adds measurable value relative to the training-free baseline, but not beyond the strongest analytical references. Hybrids that correct a closed-form prior significantly outperform the baseline (+0.103 and +0.130,

each on 11 of 12 folds, also when its articulation defect is corrected), but they do not differ from their own base learners, and given the direct-dependent count as their prior, they reproduce that count. Learned models transfer zero-shot to five stylized system models better than the training-free baseline ($\rho \approx 0.81$ vs. 0.53), but dependency counts rank even higher, and the ordering among active components remains weak. All three simulators are dominated by first-order propagation by construction, and the learners were trained on eleven synthetic architectures in one fixed configuration. Each simulator is closely tracked by a low-order structural ranking, so a regime in which learning could exceed an analytical ranking aligned with the simulator did not arise in this benchmark; whether it does under non-first-order propagation is not tested here (Section 7.2). In practice, the results favor training-free dependency analysis for routine use: it needs no training and runs in milliseconds. On I^* , recovering 80% of the critical set requires reviewing about 40–45% of Applications, so rankings suit prioritized review rather than blocking gates (Section 7.3); on I_{dyn} , the rate-weighted reference reaches 81% recall within a 20% review budget.

1.4. Research Questions

Knowing when learning is unnecessary is as useful as knowing when it succeeds. Because the three simulators studied here are dominated by first-order propagation by construction, this study tests whether learning exceeds aligned analytical truncations of such simulators; it cannot characterize the regimes in which learning would exceed them, which require simulators with non-first-order mechanisms (Section 7.5). Software engineering research has repeatedly shown that simple metrics can match more complex models when the right information is available [24, 28, 29, 30]. We therefore evaluate learned models against analytical rankings derived from the same dependencies:

- **RQ1 (Ranking accuracy):** *How accurately do analytical baselines, hybrid models, and learned approaches rank components by cascading-failure impact on unseen architectures, and can learned models exceed analytical truncations of the simulator’s propagation rule?*
- **RQ2 (Sources of predictive performance):** *At matched model capacity, which factors contribute most to ranking performance: the dependency representation, degree information, relation typing, QoS information, or the model family?*
- **RQ3 (Transfer):** *How well do learned models trained on synthetic architectures transfer zero-shot to stylized models of five open-source systems, and does their ordering of components that actually propagate failures hold?*
- **RQ4 (Cost):** *What do analytical and learned approaches cost at CI/CD time in latency and estimated energy, and how does this compare with running each simulator directly?*

The evaluation follows an analysis plan pre-specified in the replication repository. Only its two co-primary contrasts are confirmatory, and both were null; the headline findings above are registered secondary or exploratory (Section 5.3).

1.5. Contributions

1. **Dependency derivation for publish–subscribe architectures** (Section 3): typed rules that derive explicit dependency graphs from declarative architecture specifications. The Application–Library projection evaluated here (Rules 1 and 5) extends the Application-level rule of [31] with library-mediated dependencies; four infrastructure rules are formalized but not evaluated.
2. **An evaluation guideline for simulator-labeled benchmarks** (Section 4.4): an order- k reference criterion, truncating a simulator’s own computation, against which learned rankers are compared to separate genuine cascading prediction from first-order topological rule restatement.
3. **A controlled comparison of analytical, hybrid, and learned rankers** (Sections 6.1–6.3) that separates the effects of representation, degree information, relation typing, QoS information and model family.
4. **Analytical and learned approximations of a queue-flow simulator** (Section 6.1): a rate-weighted first-order approximation of a queue-flow simulator (Eq. (7)), compared with learned approximations trained on its labels.

109 **5. A reproducible benchmark, cost analysis and practitioner guidance** (Sections 5.1, 6.4 and 7.3)
110 over seventeen architectures and three simulators.

111 A previous conference paper [31] introduced the publish–subscribe multigraph, the Application-level
112 subscriber-to-publisher dependency (Rule 1 of Table 2) and a closed-form betweenness–articulation score
113 similar to the training-free baseline (Eq. (5)), validated against a reachability-loss simulation. Everything
114 else here is new: the Library and infrastructure rules, QoS weighting, the reference criterion, the learned and
115 hybrid rankers, the simulator approximations, leave-one-scenario-out (LOSO) cross-validation and zero-shot
116 evaluation, the second and third simulators, the matched controls and the cost analysis. Under this broader
117 evaluation, the closed-form score family of [31] is the weakest analytical ranker, and the present results
118 supersede that paper’s implicit recommendation of it.

119 Section 2 reviews related work, Sections 3 and 4 present the model and the rankers, Sections 5 and 6 the
120 evaluation, Section 7 the discussion and threats, and Section 8 concludes.

121 **2. Related Work**

122 *2.1. Dependability and Failure-Impact Analysis of Distributed Systems*

123 Runtime dependability methods such as broker clustering, backpressure, autoscaling, failover and chaos
124 engineering [17] require a running cluster and significant computing resources, a concern in green software
125 engineering [20, 21, 22]. Pre-production methods such as service-level fault injection [18] and microservice
126 dependency localization with PageRank (MicroRank [19]) are lighter, and benchmark platforms with cataloged,
127 replayable faults such as Train-Ticket [32] measure failure impact on a running deployment. Architecture-
128 based reliability prediction has a long history, from Cheung’s Markov model of control transfer between
129 components [33] to Yacoub and Ammar [23], who combined component dependency graphs with complexity
130 and failure severity to rank components by risk; state, path and additive reliability models [34] and the
131 Palladio Component Model [35] are other classic examples.

132 Error propagation has also been analyzed at the architecture level [36, 37, 38, 39], and safety-critical
133 practice uses model-based failure analysis such as HiP-HOPS [40] and the AADL Error Model Annex [41].
134 These frameworks derive propagation paths from structural descriptions, as SaG’s dependency projection does,
135 but they need failure-mode annotations, propagation probabilities or operational profiles, which deployment
136 manifests do not carry.

137 Telemetry-based methods localize faults in running microservices, among them Seer [42], Sage [43],
138 MicroRCA [44] and GNN-based root-cause frameworks such as Eadro [45], reviewed by Zhang et al. [46].
139 All need data from a live system; SaG works before any code is run. Production traces show that real
140 dependency graphs are large and dynamic [47], and that metastable failures are sustained by load feedback
141 rather than by the structure that failed [48, 49], which a structural ranking cannot see. SaG’s focus is close
142 to change impact analysis [50] and microservice dependency management [51], and ranking by impact relates
143 to learning-to-rank defect prediction [52].

144 Failure-impact rankings are often evaluated against a simulator rather than observed outages, so a ranking
145 may agree with a simulator because it encodes the simulator’s propagation rule rather than because it
146 predicts its behavior. The concern is not new: simulation methodology treats a model used as ground truth
147 as something to be validated [53], simple neighborhood counts track short-range spreading processes [54],
148 and learned models exploit shortcuts that encode the labeling rule [55, 56]. What we add is an operational
149 check for simulator-labeled ranking benchmarks (Section 4.4).

150 *2.2. Static Analysis and Architectural Dependency Metrics*

151 Static code analysis tools such as SonarQube [15] measure complexity [57], cohesion and coupling [16]
152 within individual services to identify defect-prone modules [58, 59]. In defect prediction, Zimmermann and
153 Nagappan [24] applied network analysis to dependency graphs, and Premraj and Herzig’s replication [28]
154 found that simple code and coupling metrics often perform as well. Such analysis cannot see inter-service
155 messaging, broker saturation or cross-host propagation. Architecture recovery tools rebuild system structure

from code [60], including microservice architecture reconstruction tools such as MicroART [61], and HAROS [62] and ROSDiscover [63] recover the run-time architecture of ROS systems, the kind of extraction SaG would need on real systems. Architecture-level coupling metrics count a service’s consumers: afferent coupling [64], the Absolute Importance of the Service [65] and service fan-in [66]. In publish–subscribe systems this count is a typed two-hop query (publisher to topic to subscriber), which SaG’s dependency rules define; because it matches the reachability oracle’s first propagation wave, this paper benchmarks it both as an established structural baseline and as an analytical reference (Section 4.4). Such analyses support the detection of anti-patterns [67, 68] and architectural technical debt [69] in CI/CD [70].

2.3. Graph Learning on Dependency Structures

Closed-form indices [71, 72, 73, 74], including this study’s training-free baseline, and cascade models of network robustness [75, 8, 9] assume homogeneous, usually undirected graphs, so a single untyped score mixes topics, libraries and hosts. Learned node-importance methods such as FINDER [25], DrBC [26] and PowerGraph [76], and learned approximations of betweenness and closeness [77], also assume homogeneity; GENI [78] estimates node importance on heterogeneous knowledge graphs. Homogeneous GNNs such as GAT [79] ignore relation identity unless it is a feature, and softmax attention averages over neighbors and cannot count them, whereas sum aggregation can [80, 81]; message-passing GNNs are limited in counting substructures without explicit features [82]. Because the main reference here is a neighbor count, this difference matters (Section 6.2). Heterogeneous GNNs such as HGT [83] learn relation-specific transformations, but Lv et al. [84] found that they often fail to beat well-tuned homogeneous baselines. GNN rankings are also sensitive to splits and tuning [85], and label propagation with shallow models can outperform GNNs [86]. Khodabandeh et al. [27] use graph attention on microservice call graphs to predict upcoming interactions; we predict the effect of removing a node from a declared topology. Learned performance models such as DeepPerf [87] likewise approximate an expensive measurement or simulation, connecting to classic simulation metamodeling [88], and performance engineering has long combined analytical models with machine learning, using the analytical model where it is accurate and learning only its residual [89]; SaG’s hybrids and its learners started from Eq. (7) follow that design.

2.4. Positioning of This Study

This study builds on findings that simple or tuned shallow methods can match deep learning at a fraction of its cost [29, 30, 90, 91] and that network measures add little over simple coupling metrics [24, 28]. It differs from prior work in three ways: it derives explicit dependency graphs from publish–subscribe deployment manifests rather than starting from an existing architectural graph; it evaluates analytical, hybrid and learned rankers on the same representation and asks where their performance originates; and it applies an operational reference criterion to separate rankings that restate a simulator’s rule from those that predict its outcomes. Rather than asking whether graph learning can rank cascade impact, it asks whether learning adds measurable value beyond analytical baselines derived directly from the explicit dependency structure.

3. The Software-as-a-Graph (SaG) Architectural Model

Figure 1 shows the SaG framework. The front end reads Architecture-as-Code manifests, builds a typed multigraph, derives a logical dependency layer that makes failure paths explicit, and extracts typed node properties, so that analytical, hybrid and learned rankings are computed from the same representation and can be compared under identical dependency information.

3.1. Multigraph Definition

A distributed system is described as a typed, weighted, directed multigraph

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E), \quad (1)$$

where $V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{host}} \cup V_{\text{lib}}$ holds the five entity types of Table 1, τ_V and τ_E assign entity and relation types, and $w_V, w_E \in (0, 1]$ weight entity criticality and connection strength. An

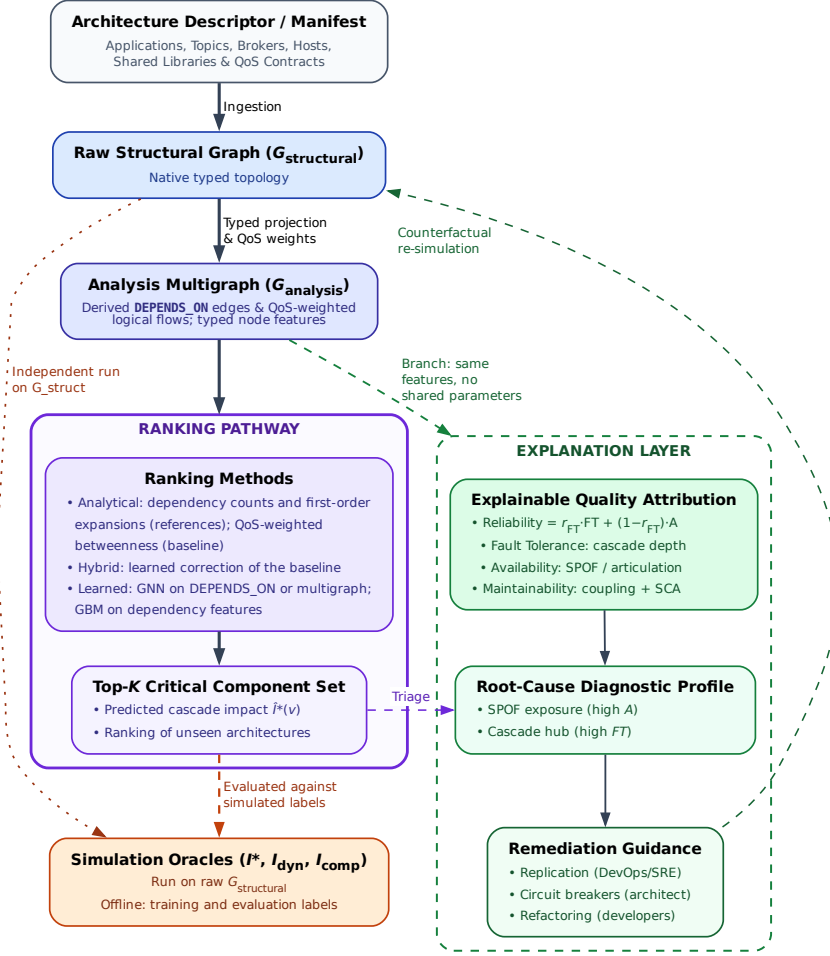


Figure 1: End-to-end architecture of the SaG framework. The ranking pathway runs down the center: manifest ingestion, typed structural multigraph ($G_{\text{structural}}$), DEPENDS_ON logical dependency projection (G_{analysis}) with typed node properties, the ranking methods (analytical, hybrid, and learned; Figure 3), and the ranked critical set. The dashed edge marks the simulation oracles, which operate strictly on $G_{\text{structural}}$, provide training and evaluation labels offline, and do not participate in inference. The explanation layer is a conceptual proposal outside the empirical evaluation of this study (Supplementary Section S26).

Application's w_V is the power mean ($p = 3$) of the QoS weights $w(t)$ (Section 3.2) of its topics. A Library ℓ takes the largest weight among its topics $T(\ell)$ and consuming Applications $C(\ell)$, amplified by its fan-out: $w_V(\ell) = \min(1, \max(\{w(t)\}_{t \in T(\ell)} \cup \{w_V(a)\}_{a \in C(\ell)}) \cdot (1 + 0.15 \log_2(1 + |C(\ell)|)))$. The raw multigraph is $G_{\text{structural}}$; the analysis graph G_{analysis} adds the derived DEPENDS_ON edges (Section 3.3) and the node properties computed from them (Section 3.5). Supplementary Section S1 lists the notation.

Scope. The evaluation ranks *Applications* only: microservice containers, service agents or communicating processes, the developer-authored code that changes in CI/CD. Brokers, Topics, Hosts and Libraries are modeled because failures propagate through them, but ranking them is a provisioning question rather than a commit-time one (Section 7.5).

3.2. Quality-of-Service Link Weighting

A **RELIABLE** topic with **TRANSIENT_LOCAL** durability couples services more strongly than a **BEST_EFFORT** telemetry stream. Each topic t therefore has a weight $w(t) \in (0, 1]$ combining its QoS policies (reliability, durability, priority) with payload size and publication frequency. The policy weights come from an Analytic Hierarchy Process (AHP) [92] pairwise-comparison matrix defined independently of any target vector ($CR = 0.016$; Supplementary Section S6). Declared QoS policies turned out to carry no measurable signal on

Table 1: Entity types and structural edge types in the SaG model.

Panel A: Entity Types ($\mathcal{T}_V$)		
Entity Type	Architectural Role	Concrete System Examples
Application (V_{app})	Process producing/consuming messages	ROS 2 node, Kafka microservice, MQTT client
Broker (V_{broker})	Message routing and queuing intermediary	RabbitMQ exchange, Mosquitto, EMQX broker
Topic (V_{topic})	Named logical communication channel	/sensor/lidar, orders.payment.completed
Execution Host (V_{host})	Physical or virtualized execution environment	Bare-metal server, Kubernetes worker, Cloud VM
Library (V_{lib})	Shared software package or runtime dependency	Kafka client, OpenCV, Protobuf runtime
Panel B: Structural Edge Types ($\mathcal{T}_E$)		
Structural Edge	Direction	Semantic Meaning
PUBLISHES_TO / SUBSCRIBES_TO	App/Library $\rightarrow$ Topic	Component publishes to / consumes from topic
ROUTES	Broker $\rightarrow$ Topic	Broker manages and routes topic traffic
RUNS_ON	App/Broker $\rightarrow$ Host	Process is hosted on host
CONNECTS_TO	Host $\rightarrow$ Host	Network link between hosts
USES	App $\rightarrow$ Library	Application links to shared library

either the reachability or the queue-flow simulator (Sections 6.1 and 6.2); for I^* this is largely fixed by design, because it barely reads QoS (Section 4.3). The dependency counts reported in this paper are unweighted.

3.3. Logical Dependency Projection (DEPENDS_ON)

Structural edges do not show how failures spread: a subscriber depends on a publisher, but no edge joins them. SaG therefore derives DEPENDS_ON edges from the dependent to the dependency (if the target fails, the source is affected) using the rules of Table 2. The resulting graph is the common input of the analytical, hybrid and learned rankers.

Table 2: The DEPENDS_ON logical dependency projection rules. Rules 1 and 5 define the operational Application–Library projection evaluated in this study; $\dagger$ rules define broader infrastructure relations formalized for system-level completeness but unexercised by the Application-level evaluation.

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
1	app_to_app	Subscriber $\rightarrow$ Publisher (via shared Topic)	$1 - \prod_{t \in T} (1 - w(t))$
2 $\dagger$	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3 $\dagger$	host_to_host	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	$\max_{u \in \text{hosted}(h_1), v \in \text{hosted}(h_2)} w_E(u \rightarrow v)$
4 $\dagger$	host_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	$\max_{u \in \text{hosted}(h)} w_E(u \rightarrow b)$
5	app_to_lib	Application $\rightarrow$ Shared Library it USES	$H(w_V(\text{app}), w_V(\text{lib}))$
6 $\dagger$	broker_to_broker	Broker $\leftrightarrow$ Broker (shared fault-domain colocation, symmetric)	$w_V(\text{host})$

Rules 1 and 2 combine the topics T connecting a pair by probabilistic union [93], so multiple failure paths strengthen the connection; Rule 5 uses the harmonic mean $H(x, y) = 2xy/(x + y)$ [94]. Rule 1 captures sequential cascades (a failed publisher starves its subscribers) and Rule 5 simultaneous blasts (a crashed library fails every Application using it); neither is stated explicitly in a manifest. Rules 2, 3, 4 and 6 are formalized for architectural completeness across host and broker tiers, but are not exercised in the present Application-level evaluation.

Remark 1 (the dependency count is a typed two-hop count). Let G_{flow} be the Application–Library projection (Rules 1 and 5) and v an Application. By construction,

$$\text{InDeg}(v) = |\{u \neq v : \exists t \in V_{\text{topic}}, (v, t) \in \text{PUBLISHES_TO} \wedge (u, t) \in \text{SUBSCRIBES_TO}\}|, \quad (2)$$

because Rule 5 edges end at Libraries, so every edge into an Application is a Rule 1 edge, and parallel topics between a pair collapse into one edge. The right side is publish–subscribe afferent coupling [64, 65], a typed two-hop query on the raw multigraph; the projection defines which query to use (the equality holds exactly on all seventeen corpus graphs). The derivation adds value beyond this count: including Rule 5 raises the agreement of transitive reach with I^* by +0.058 (9 of 12 folds, Holm $p = 0.0068$; Section 6.1). Because I^* ’s first propagation wave matches this set (Section 4.4), InDeg and its transitive version Reach serve as both established structural baselines and first-order analytical references. The in-degree *feature* the learners receive is computed across all incoming edges on G_{analysis} (including incoming library links) and normalized by graph size, whereas the InDeg reference counts raw incoming Rule 1 publisher-subscriber edges between

Applications; the two differ for 940 of the 1,321 Applications, with per-fold rank agreement $\rho = 0.55\text{--}1.00$ (Supplementary Section S13).

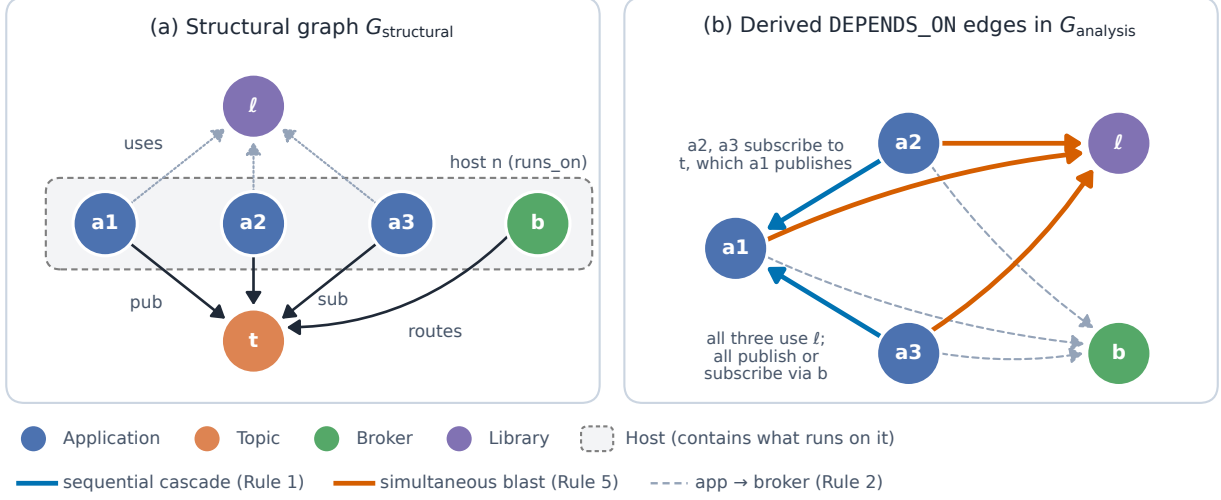


Figure 2: Running example. (a) Three applications share topic t (routed by broker b) and library ℓ , and all run on host n . No structural edge joins two applications. (b) The derived DEPENDS_ON edges make the hidden dependencies explicit: subscribers a_2, a_3 depend on publisher a_1 (Rule 1), all applications depend on ℓ (Rule 5), and each application depends on the broker (Rule 2; drawn for completeness, but not part of the Application–Library projection evaluated here). Simulation oracles run on view (a); the analytical rankings and the -P learners read view (b), and the raw-graph learners pass messages over view (a) with node features computed from (b).

3.4. Dual Graph Views

The **structural graph** $G_{\text{structural}}$ is the raw deployment topology; the **analysis graph** G_{analysis} adds the DEPENDS_ON edges and code metrics (Figure 2). Every ranker’s node features are computed on G_{analysis} , and no ranker reads simulator output. The learners pass messages either over $G_{\text{structural}}$ or over the DEPENDS_ON projection (the -P models), and the difference matters: on $G_{\text{structural}}$ every relation points away from Applications (PUBLISHES_TO , RUNS_ON , USES), so forward message passing never reaches them and forward GNNs collapse into per-node models over precomputed features. On the DEPENDS_ON graph each Application receives messages from its dependents. Because the views also differ in edge direction, Section 6.2 adds a raw-graph learner that passes every message in both directions, separating dependency semantics from direction.

3.5. Typed Node Feature Encoding

All five entity types share 18 normalized metrics computed on G_{analysis} , including PageRank, reverse PageRank, betweenness, closeness, in- and out-degree, an articulation score, QoS and dependency weights, multi-path coupling, fan-out criticality and the Connectivity Degradation Index (CDI); type-specific blocks extend the vector to 19–25 dimensions (Applications and Libraries add a code-quality penalty and its four static-code inputs). The full schema is in Supplementary Section S13. Four centralities use QoS-weighted edges, so the QoS-off condition is not entirely free of QoS. Several features compute part of what the reachability simulator computes: in-degree and its QoS-weighted version w_{in} are close to InDeg , CDI is a removal-and-reachability computation, and multi-path coupling, reverse PageRank, fan-out criticality and the articulation score are also aligned with it. Every published learner reads these *oracle-aligned* features and so inherits part of the oracle’s circularity; Section 6.2 reports learners with all of them removed.

4. Ranking Methods, Simulation Oracles, and the Reference Criterion

The study compares analytical rankings derived directly from dependency structure (the training-free baseline **Topo-QoS** and the references of Section 5.2), hybrid methods that correct an analytical prior with a learned model, and learned methods: graph neural networks on the typed multigraph or the dependency graph, and gradient-boosted models on per-Application features (Figure 3). Full hyperparameters and training commands are in the replication repository’s experiment pages.

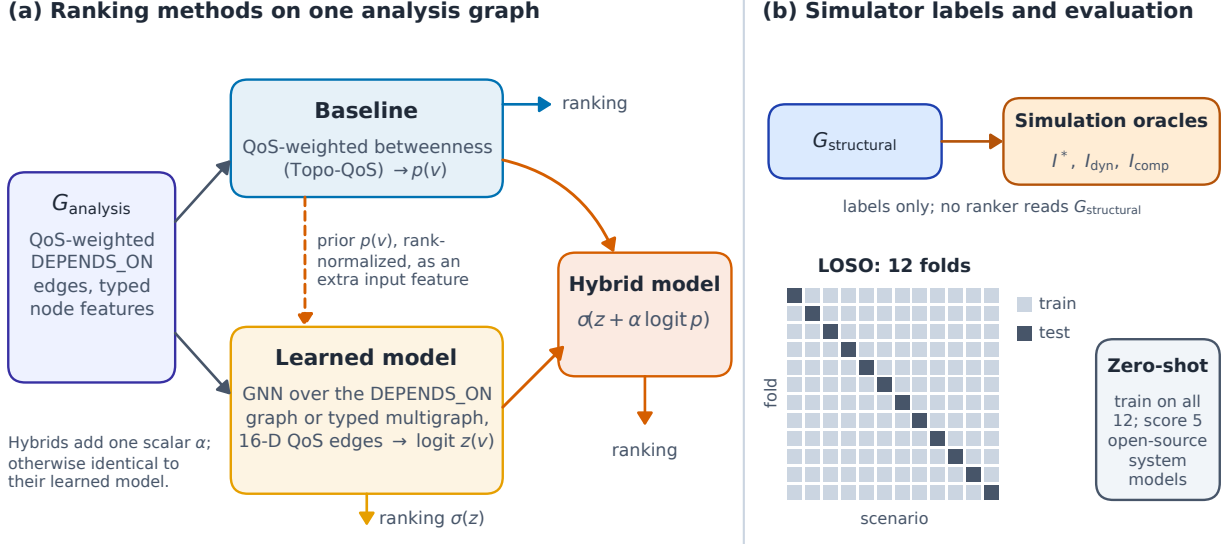


Figure 3: (a) SaG’s ranking methods read the analysis graph. The training-free baseline scores QoS-weighted betweenness; $p(v)$ is that **Topo-QoS** score, rank-normalized to $[0, 1]$ within the graph; the learned model outputs a logit $z(v)$. A hybrid model gives the learned model $p(v)$ as an extra input feature and adds a learned correction to it on the logit scale, $\sigma(z + \alpha \text{logit } p)$, with one learnable scalar α . (b) Labels come from simulation oracles on the structural graph, which no ranker reads. We evaluate the methods by leave-one-scenario-out cross-validation over twelve synthetic architectures and zero-shot on five open-source system models.

4.1. Heterogeneous Graph Transformer and Attention Networks

The main learned ranker, **HGT-QoS**, is a three-layer Heterogeneous Graph Transformer [83] (PyTorch Geometric [95]; hidden dimension 64, four heads) with entity-specific input projections and type-parameterized attention per meta-relation (Supplementary Section S2). It passes messages over its input graph and its transpose: $G_{\text{structural}}$ for **HGT-QoS**, the **DEPENDS_ON** projection for **HGT-P-QoS** (width 100, keeping parameters within 1%). The homogeneous Graph Attention Networks (**GAT-QoS**, and **GAT-P-QoS** on the dependency graph) use three **GATConv** layers with four heads at a matched parameter budget and pass messages forward only, except the direction control **GAT-QoS-R**, which also passes every edge in reverse. Each edge carries a 16-dimensional vector (coupling weight, path count, a one-hot relation encoding and seven middleware QoS parameters) that is projected and added before attention. Because the vector encodes the relation, the “untyped” **GAT** receives relation identity as a feature, but this matters only where messages reach Applications (Section 3.4); on the dependency projection every edge carries the same relation.

4.2. Prediction Head and Training Objective

A composite head predicts the simulated cascade impact, with an auxiliary head $\hat{a}_1$ regressing on I_R^* , the reliability component the same simulator run records. The training objective combines regression with listwise and margin-ranking terms:

$$\mathcal{L} = \text{MSE}(\hat{I}^*, I^*) + 0.5 \cdot \text{MSE}(\hat{a}_1, I_R^*) + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}}, \quad (3)$$

where $\mathcal{L}_{\text{rank}}$ is the Listwise Maximum Likelihood Estimation (ListMLE) loss [96] over the permutation π induced by the simulated labels,

$$\mathcal{L}_{\text{rank}} = -\frac{1}{N} \sum_{i=1}^N \left(\hat{s}_{\pi_i} - \log \sum_{j=i}^N \exp(\hat{s}_{\pi_j}) \right), \quad (4)$$

and $\mathcal{L}_{\text{pairwise}}$ is a margin-ranking loss with $\gamma = 0.05$. Tied labels (about 31% of Applications have $I^* = 0$) enter π in the generator’s creation order; a node-order permutation control tests whether this leaks into the results (Section 6.2).

Models are trained with AdamW ($\eta = 3 \times 10^{-4}$, weight decay 10^{-4}) and cosine warm restarts for up to 300 epochs, with early stopping (patience 60) on a 20% node-level validation split of the largest training scenario of each fold, over five fixed seeds. Selecting a 20% node-level split within the largest training scenario isolates early stopping from the held-out test scenario (preserving scenario-level independence under leave-one-scenario-out cross-validation) while avoiding the prohibitive computational cost of a nested scenario split across 300 epochs and five seeds. All reported evaluations are inductive under leave-one-scenario-out cross-validation. Hyperparameters and loss coefficients were fixed before the first sweep and kept across all experiments, rather than selected by the registered nested procedure (Section 5.3).

4.3. Simulation Oracles

Labels come from three failure simulators (“oracles”) that run on the raw multigraph $G_{\text{structural}}$ and define different notions of cascade impact rather than observations of real failures:

- **Reachability cascade oracle (I^* , primary):** crashes component v , propagates outages through dependent topics, brokers and links by breadth-first traversal, and computes the mean fractional feed loss across intact subscribers, with each topic’s loss scaled by a declared, uncalibrated QoS severity ladder ($\times 1.2$ for RELIABLE, $\times 1.15$ or $\times 1.05$ for high or medium priority). A subscriber whose feed loss exceeds the propagation threshold (0.2) fails with a probability that is one in the first wave and decays by 0.15 per later wave (floor 0.25); the label averages five seeds of this stochastic propagation (pseudocode in Supplementary Section S11). Disabling QoS scaling leaves the Application ordering largely intact ($\rho = 0.965$ across the twelve folds), and substituting durability-aware rescaling moves it less still ($\rho = 0.977$): I^* is predominantly a topological reachability metric.
- **Queue-flow oracle (I_{dyn}):** a discrete-event SimPy [97] simulation of message rates, bounded subscriber queues and service contention that measures the drop in delivered message rate for surviving consumers, using declared publication rates and QoS contracts (history depth, reliability, deadlines, lifespan, priority) that I^* ignores. It does not read declared payload sizes (every message has the same size), model brokers or network links, block publishers on full queues, or retry; a consumer that loses its inputs keeps publishing. Its first-order effect, consumers losing a failed publisher’s messages, is therefore again subscriber loss, and by construction it propagates no failure beyond one hop. It labels all 1,321 Applications (five seeds) in 12.7 CPU-hours, against seconds for I^* , and agrees with I^* at a mean $\rho = 0.711$. A single I_{dyn} run agrees with another at $\rho = 0.43$ – 0.96 per fold; the five-seed mean used as the label has an estimated reliability of $r = 0.79$ – 0.99 (Spearman–Brown), so no ranker can agree with it by more than $\sqrt{r} \approx 0.89$ – 0.996 per fold, and every ranker in Table 6 stays below r , and therefore below $\sqrt{r}$, on every fold. While the Spearman–Brown formula is an approximation when applied to rank correlations, it establishes the repeatable signal envelope, indicating substantial fold-level headroom ($\sqrt{r_f} - \rho_f \geq 0.05$). Importantly, the GNN evaluated on I_{dyn} (GAT-P-QoS- $\rightarrow$ dyn) received only static topological and QoS edge features without declared publication rates or payloads, whereas LightGBM and the first-order reference (Eq. (7)) received declared publication rates directly.
- **Composite multi-criteria oracle (I_{comp}):** the composite multi-criteria failure simulator, run on all Applications, scores each removal as $I_{\text{comp}}(v) = 0.35 \text{RL}(v) + 0.25 \text{FR}(v) + 0.25 \text{TL}(v) + 0.15 \text{FD}(v)$ over reachability loss, fragmentation, throughput loss and flow disruption. The weights are declared, not elicited, so I_{comp} is a multi-objective stress test rather than a measure of subscriber loss; it is never a training label.

4.4. Input-Label Separation and the Reference Criterion

No simulation output is used as an input (a regression test enforces this). Procedural separation does not remove construct overlap, however: a ranking may agree with a simulator because it reproduces the simulator’s rule rather than because it infers impact from independent evidence.

The reference criterion. Let an oracle O compute the impact of removing v by propagating the failure in waves over its inputs, and let $T_k(O)$ be the same computation stopped after wave k . A ranking R is an *order- k reference* for O if R equals $T_k(O)$, computed from the inputs O reads, after at most the following simplifications: (S1) the expectation in place of O ’s seed-dependent stochastic propagation; (S2) omitting a severity rescaling that O applies to each loss term; (S3) omitting the mechanisms O adds on top of propagation (queueing, drops, deadlines); (S4) uniform weights in place of per-component rates or normalizations; and (S5) the support of a wave (the number of affected components) in place of its weighted loss. Simplifications S1–S5 explicitly isolate the deterministic, low-order topological skeleton of the simulation oracle from stochastic sampling variance (S1), arbitrary severity scales (S2), higher-order queueing dynamics (S3), and continuous rate parameterizations (S4, S5). No other simplification is admitted, and no parameter of R may be fitted to the output of O . Whether a ranking qualifies is therefore decided by structural derivation from the simulator’s generative rules, not by its empirical score. Crucially, canonical graph centrality measures (such as betweenness centrality, PageRank, and eigenvector centrality) fail this criterion because they evaluate all-pairs shortest paths or global random-walk stationary distributions rather than truncated failure propagation waves; like learned GNNs, they are classified as *predictors*, confirming that the criterion is structurally grounded rather than post hoc baseline selection. For I^* , where direct dependents (C_a) dominate outcomes, the first-order expansion (Eq. (6)) is $T_1(I^*)$ under S1, S2 and S4 (equal publication rates, and no division of a subscriber’s loss by its number of feeds), **InDeg** is its support (S5; Remark 1), and **Reach** is the support of the untruncated cascade. The reference criterion thus makes explicit that I^* fundamentally evaluates publish–subscribe afferent coupling under stochastic perturbation. For I_{dyn} , the rate-weighted expansion (Eq. (7)) is $T_1(I_{\text{dyn}})$ under S3 and S4, the delivered-rate loss after one wave without queueing. **Topo-QoS**, the hybrids and every learned ranker fail the definition and are *predictors*. For I_{comp} no ranking is a truncation, but its fragmentation and flow terms respond to removing high-degree and high-betweenness nodes, so **Topo-QoS** and raw degree are *term-aligned* with it and read with the same caution.

A reference measures how much of O its own rule recovers, a property of the oracle rather than predictive skill. Comparisons between a predictor and a reference are therefore descriptive anchors (how close a predictor comes to the oracle’s own first-order rule), not registered contrasts, and the statement that no learned model exceeded a reference is descriptive; the one exception, the exploratory paired comparison of Eq. (7) with the learned queue-flow approximation, is flagged where it appears (Section 6.1). We defined the criterion post hoc, after simple dependency counts performed strongly; reclassifying them changed no number, but the criterion has not been applied to a corpus generated after the analysis was frozen (Section 7.5).

Learned methods are not references by this definition, but they are not less circular: they are fitted to the oracle they are scored against and read oracle-aligned features (Section 3.5). We therefore also report learners with those features removed and learners’ agreement with I_{dyn} after I^* or the first-order term is partialled out (Section 6). Neither replaces validation against real failures (Section 7.4).

5. Experimental Setup

The design asks whether ranking performance originates from the dependency representation or from the learning algorithm operating on it. If it arises from the representation, analytical and learned rankers built on that representation should perform similarly; if learning adds information, learned rankers should outperform analytical alternatives built on the same dependencies.

5.1. Corpus and Replication Package

The corpus includes 2,812 components from seventeen architectures (Table 3). Twelve synthetic topologies form the LOSO folds, covering autonomous vehicles, financial trading, healthcare, industrial SCADA, smart-city IoT, telecom RAN, logistics, gaming, microservices, enterprise integration and air-traffic management.

All twelve come from one generator family, so LOSO tests transfer across its parameter settings, not across unrelated industrial systems; each topology regenerates exactly from its configuration, checked in CI against a SHA-256 manifest.

Table 3: Evaluation corpus. The twelve synthetic topologies are the LOSO folds; we exclude the five open-source system models from all training and use them only for zero-shot transfer (Section 6.3). Counts are read from the committed topology files and verified in CI; Supplementary Section S15 gives per-scenario composition.

Dataset	$ V $	$ V_{\text{app}} $	Topics	Brokers	Hosts	Libs	$ E $
Synthetic evaluation scenarios (12)	2,461	1,321	615	65	202	258	10,918
Open-source system models (5)	351	141	120	16	32	42	700
Total	2,812	1,462	735	81	234	300	11,618

The five open-source system models (Autware.universe on ROS 2, EdgeX Foundry, Home Assistant, and meshes based on Online Boutique and Train-Ticket) were each hand-authored by the first author from public architectural documentation, not extracted from source code or manifests, and no second modeler re-derived any of them. Brokers, QoS profiles, code metrics and host specifications are partly assumed. Online Boutique and Train-Ticket are RPC systems re-expressed as event-driven publish-subscribe meshes: synchronous calls become one-way event, command or request topics on assumed brokers, with no reply topics, so the models drop request-reply coupling, timeouts, thread-pool exhaustion and synchronous backpressure (Supplementary Section S15). No topology in this study therefore comes from a real deployment manifest, and RQ3 tests transfer to stylized single-modeler models, not to production systems.

All datasets, harnesses, checkpoints and result artifacts are on Zenodo (see Data Availability), and the public repository documents each experiment’s protocol, hyperparameters, `make` target and artifacts (<https://github.com/onuralpyigit/software-as-a-graph/tree/56d9bff8ae9583ee9df1d270f0650a3a7c3239e8/docs/research/jss/experiments>). Independent verification of all experimental scripts and data pipelines was conducted through automated regression test suites and an immutable 1,817-figure reconciliation suite ensuring exact provenance.

5.2. Rankers and References

Table 4 groups the rankers as analytical, hybrid and learned; further training-free baselines and variants are in Supplementary Sections S43 and S34. GAT and GAT-QoS are untyped GATs at HGT’s parameter budget (matched within $\pm 1.7\%$ across the 2×2 design, ranging from 429,992 to 437,496 parameters; Table 4), giving a 2×2 of typing and QoS channel. The “-P” models read the Application-Library `DEPENDS_ON` graph with node features and labels identical to the raw-multigraph models. QoS-off arms set edge weights to 1 and zero the three QoS node columns (w , w_{in} , w_{out}).

Training-free baseline. Topo-QoS is computed on the Application-Library `DEPENDS_ON` graph (Rules 1 and 5):

$$\text{Topo-QoS}(v) = 0.6 \cdot \text{BT}_w(v) + 0.4 \cdot \text{AP}(v), \quad (5)$$

where BT_w is betweenness over edge distances $d(e) = 1/(w(e) + 10^{-6})$ and AP flags articulation points. Due to an implementation defect, the AP term is zero for every node: the scorer reads the articulation flag under a key (`ap_c_score`) that the cached metrics do not contain, and falls back to zero, so every table reports QoS-weighted betweenness alone. The defect slightly favors the baseline: restoring the term lowers Topo-QoS from 0.553 to 0.533, so it does not bias comparisons toward the learned models or the references. The hybrids’ prior $p(v)$ inherits the defect. The registered contrasts stay against the registered value, because replacing a comparator after its contrasts were registered would be an undeclared deviation; Table 5 reports the corrected value beside it, and hybrids retrained on the corrected prior are compared with it in Section 6.1 (Table 7, F9). The comparator was fixed when the analysis plan was written, before the dependency counts had been analyzed, and continues the score family of [31]; in hindsight, afferent coupling would have been the natural comparator, and it is reported beside the baseline as a reference.

Table 4: Rankers and references reported in the main text. **-QoS**: QoS-weighted distances (**Topo-QoS**) or the 16-D QoS edge vector plus three QoS node columns (GNNs); **-P**: trained on the derived dependency graph; **Hybrid-X**: model X corrected by the **Topo-QoS** prior; **→dyn**: trained on I_{dyn} labels. Unsuffixed GNNs are raw-multigraph controls. Params: trainable parameters. References restate a simulator’s rule and carry no contrast (Section 4.4); control arms are in Section 6.2, where further suffixes mark the change made to an arm: **-R** reverse edges, **-deg/-min/-const** feature removal, **-AP** corrected prior, **-perm** permuted node order. [¶]Articulation term zero by an implementation defect (see text).

Ranker	Graph read	Typing / edge input	Params
<i>Analytical: training-free baseline (registered comparator)</i>			
Topo-QoS [¶]	App-Lib DEPENDS_ON	scalar $w(e)$; betweenness only (AP term zero; corr. 0.533)	—
<i>Analytical: references (no contrast; Section 4.4)</i>			
Analytic I^* (Eq. (6))	topic publish/subscribe sets	first-order subscriber loss	—
Rate-weighted (Eq. (7))	topic publish/subscribe sets	first-order loss weighted by declared rates	—
InDeg / Reach	App-Lib DEPENDS_ON	direct / transitive dependents	—
<i>Hybrid: learned correction of the Topo-QoS prior</i>			
Hybrid-GAT / Hybrid-HGT	$G_{\text{structural}}$ + features	as base + Topo-QoS prior	431,433 / 434,941
<i>Learned: dependency graph</i>			
GAT-P-QoS , HGT-P-QoS	App-Lib DEPENDS_ON	homogeneous / heterogeneous	429,992 / 430,680
<i>Learned: raw multigraph, typing $\times$ QoS channel at matched budget</i>			
GAT / GAT-QoS	$G_{\text{structural}}$ + features	homogeneous; none / 16-D	437,496 / 429,992
HGT / HGT-QoS	$G_{\text{structural}}$ + features	heterogeneous; 1-hot / 16-D	434,620
<i>Learned approximations of I_{dyn} (trained on I_{dyn} labels)</i>			
GBM-P-QoS→dyn	per-Application features (App-Lib DEPENDS_ON)	tabular S+Q design: 9 dependency counts + 9 QoS/rate columns	—
GAT-P-QoS→dyn	App-Lib DEPENDS_ON	homogeneous; 16-D	429,992

References. Three training-free rankings restate the reachability oracle’s propagation rule. The first is the post hoc first-order expansion of I^* (Analytic I^*):

$$\hat{I}_1^*(v) = \sum_{t \in \text{pub}(v)} \frac{|\text{sub}(t)|}{|\text{pub}(t)|}, \quad (6)$$

where $|\text{pub}(t)| \geq 1$ for all topics published by v . The second is **InDeg**(v), the number of direct dependents on the DEPENDS_ON graph (Remark 1), and the third **Reach**(v), the number of transitive dependents normalized by $|V| - 1$. For the queue-flow simulator I_{dyn} , the corresponding reference weights each term of Eq. (6) by the topic’s declared publication frequency r_t :

$$\hat{I}_{\text{dyn},1}^{\text{rate}}(v) = \sum_{t \in \text{pub}(v)} \frac{r_t}{|\text{pub}(t)|} |\text{sub}(t)|. \quad (7)$$

Here, r_t is the declared publication rate (msg/s) of topic t . It is a first-order truncation of I_{dyn} , which reads the same rates, and was evaluated after the registered analyses (exploratory). References carry no contrast against **Topo-QoS**; the one exception is the exploratory paired comparison between Eq. (7) and the learned queue-flow approximation (Section 6.1).

5.3. Metrics, Protocols and Statistics

Metrics and protocols. Every ranker is scored on the Application set V_{app} with Spearman ρ against each oracle, the active-stratum $\rho_{>0}$ over components with true impact above zero, and **Overlap@K**, the fraction of the true top- K recovered by the predicted top- K ($K \approx 20\%$ of $|V_{\text{app}}|$); **PR-AUC**, $F_1@r$ and **nDCG@10** are in Supplementary Section S17. Under LOSO, learned models train on eleven scenarios and are tested on the twelfth, for all 12 folds and five seeds, and the reported value is the mean ρ across seeds. For zero-shot transfer, models trained on all twelve scenarios are evaluated on the five system models without fine-tuning. Training-free rankers are scored on the same scenarios and labels.

Statistics. The summary ρ is the mean over folds or systems with a 95% bootstrap interval ($B = 2,000$) [98]. Contrasts use paired two-sided Wilcoxon signed-rank tests across folds [99, 100], and, because LOSO folds share ten of eleven training scenarios, every registered contrast is also tested with the Nadeau–Bengio corrected resampled t -test [101]. The Nadeau–Bengio test-to-train ratio is 1/11, one held-out scenario per eleven training scenarios; taking Applications as the unit instead changes no conclusion (for example, $p = 0.019$ vs. 0.020 for Hybrid-HGT). “Won” is the number of folds on which a ranker beats its comparator, a sign count

reported beside the paired $\Delta\rho$ and its interval, not an effect size in its own right. “Not significantly different” does not mean statistically equivalent; equivalence is claimed only where a two one-sided test at ± 0.05 passes, and where the paper says one ranker “matches” another, it means the two are not significantly different. Overlap@ K breaks ties by a deterministic but arbitrary order (NumPy’s default sort over components ordered by identifier); Figure 5 resolves ties in expectation instead. Partial Spearman correlations (Table 6) are computed per fold as the Pearson correlation of the residuals of both ranks after linear regression on the rank of the controlled ranking, then averaged over folds with the same bootstrap interval.

Analysis plan and the status of each result. The protocol is pre-specified in the replication repository (initial plan registered at commit 56d9bff8), not lodged with a third-party registry; seventeen numbered amendments were registered, none withheld. *Confirmatory*: the plan’s two co-primary contrasts (HGT-QoS and HGT vs. Topo-QoS), both null. *Registered secondary*: arms committed before they ran but after the primary null was known (the matched 2×2 , the hybrids, the reference counts, the derivation value, the full-population queue-flow labels and the control arms of Section 6.2), each Holm-corrected within its family. *Exploratory*: post hoc analyses, the dependency-graph learners’ contrasts, the reference reclassification (which changed no figure) and the rate-weighted reference with the attribution of the learned queue-flow approximation. **Plan deviation**: the registered nested hyperparameter selection was replaced by one fixed configuration, and the registered rule was run later as a sensitivity check (Section 6.2). An omnibus Holm correction across the 13 decision-bearing contrasts (Supplementary Section S27) leaves the hybrid contrasts significant (Hybrid-GAT $p_{\text{omni}} = 0.019$, Hybrid-HGT $p_{\text{omni}} = 0.041$) and the primary contrast null ($p_{\text{omni}} \geq 0.46$). The headline findings of this paper are therefore registered secondary or exploratory, not confirmatory.

Status	Contrast / Analysis	Statistical Test	Result
<i>Confirmatory</i>	HGT(-QoS) vs. Topo-QoS	Paired Wilcoxon, Nadeau-Bengio	Null ($\Delta\rho = +0.008$, $p = 0.622$)
<i>Registered secondary</i>	Factorial typing $\times$ QoS	Factorial ANOVA / Wilcoxon	Typing null (+0.003); QoS edge features +0.072
<i>Registered secondary</i>	Hybrids vs. Topo-QoS	Paired Wilcoxon, Nadeau-Bengio	Supported ($\Delta\rho = +0.072$, $p_{\text{omni}} = 0.019$)
<i>Registered secondary</i>	Reference bounds (I^* , InDeg)	Paired comparisons vs. bounds	Reference matches GNNs ($\rho = 0.772$ vs. 0.748)
<i>Registered secondary</i>	Zero-shot transfer (RQ3)	LOSO transfer to 5 systems	Preserved on aggregate; active stratum weak
<i>Exploratory</i>	Rate reference vs. GAT-P-QoS $\rightarrow$ dyn	Paired Wilcoxon	Rate reference outperforms GNN (0.830 vs. 0.598)
<i>Exploratory</i>	Post hoc reference reclassification	Structural derivation from oracle	Isolates order- k simulation truncations

6. Results

All results are reported on the Application population against the primary oracle I^* and, in Table 6, also against I_{dyn} and I_{comp} (Section 4.3). Section 5.3 gives the status of each result; per-fold values are in the experiment pages. Figure 4 summarizes the main findings.

6.1. RQ1: Ranking Accuracy

Table 5 reports every ranker against I^* (learned rankers trained on eleven scenarios and scored on the twelfth; 26 to 300 Applications per fold, K between 5 and 60). Two patterns emerge. Reading the dependency graph instead of the raw multigraph improves attention-based learners (GAT-P-QoS 0.748 vs. GAT-QoS 0.635), although the typed transformer becomes unstable on it (HGT-P-QoS 0.514). And no learned model exceeded the references derived from the same dependencies (0.764–0.808, against at most 0.748), a descriptive comparison rather than a registered contrast (Section 4.4). Because the registered baseline Topo-QoS evaluates QoS-weighted betweenness alone due to an implementation defect, the meaningful structural reference is afferent coupling (InDeg, 0.764). Weighting folds by $|V_{\text{app}}|$ keeps the top five rows but lifts Topo-QoS to 0.596, level with the raw-multigraph learned rankers (0.595–0.606).

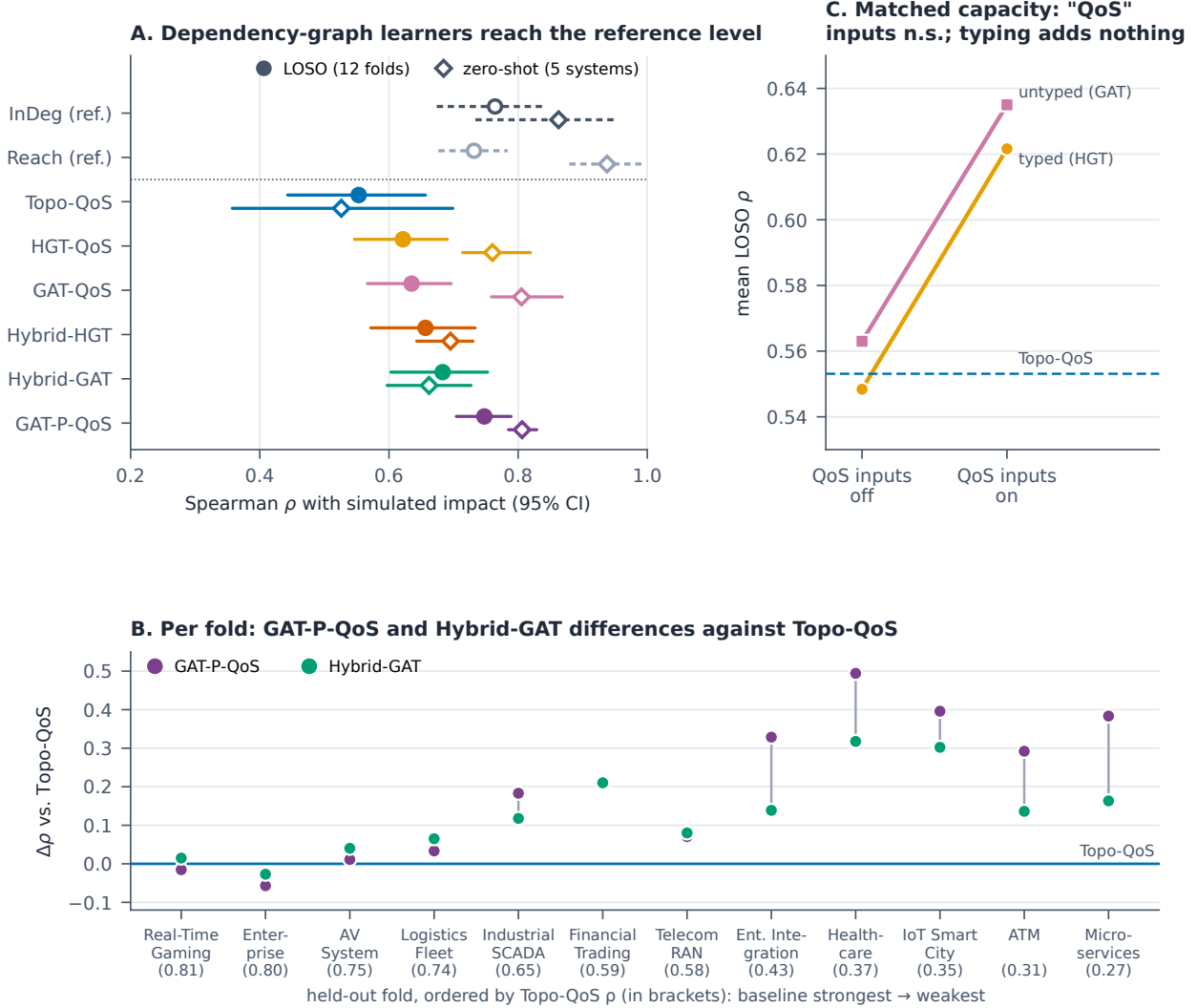


Figure 4: Main results for the Application population. (A) Mean Spearman ρ with 95% bootstrap intervals under LOSO (filled circles; Table 5) and zero-shot on five system models (open diamonds); grey hollow markers are references that restate I^* 's rule (InDeg, Reach). GAT-P-QoS is not significantly different from InDeg, though equivalence within ± 0.05 is not established (TOST $p = 0.17$). (B) Per-fold improvement of GAT-P-QoS and Hybrid-GAT over Topo-QoS. (C) Cell means of the capacity-matched 2×2 on the raw multigraph (Supplementary Table S25): the "QoS" inputs, which include a QoS-weighted in-degree column, raise both models by about 0.07 (not significant after Holm), falling to +0.030 with w_{in} held in both arms (Table 7).

The reference level. The references measure how much of I^* restating its rule recovers, not predictive skill. The first-order expansion (Eq. (6)) reaches 0.808 ($\rho_{>0} = 0.631$): direct subscriber loss is most of what I^* measures. InDeg, the support of that first wave (Remark 1), is a coarser version of it (0.764, $\rho_{>0} = 0.516$), and Reach reaches 0.732. The derivation matters for the latter: without Rule 5, Reach drops by 0.058 (9/12 folds, Holm $p = 0.0068$). Untyped raw-graph metrics do not recover the reference (total degree 0.199, reverse PageRank 0.089).

Learners on the dependency graph approach the reference. GAT-P-QoS reaches $\rho = 0.748$, 0.016 below the direct-dependent count per seed and 0.008 above it as a five-seed ensemble (0.772, Table 6). Neither difference is significant, and neither shows equivalence: the 90% interval of the per-fold difference is ± 0.076 , so a two one-sided test at the registered margin of ± 0.05 does not pass ($p = 0.17$ per seed, 0.13 for the ensemble). Both differences are also smaller than the per-fold variation that node order alone induces in this learner (about 0.044; Section 6.2). The closeness depends on the degree features (Section 6.2).

Table 5: Main results under LOSO across twelve synthetic architectures (Application population; learned rankers: five seeds, CPU sweeps). $\Delta\rho$ is paired by fold against **Topo-QoS**, with a bootstrap 95% CI and a two-sided Wilcoxon test; p_{Holm} is within the registered hybrid family; $\ddagger$ exploratory. $\rho_{>0}$: Spearman over active components ($I^* > 0$). Overlap@ K ($K \approx 20\%$) breaks ties by deterministic identifier sort; Figure 5 resolves ties at zero in expectation. References restate I^* ’s rule and carry no contrast. $\P$ Registered comparator, whose articulation term is zero because of an implementation defect (Section 5.2), so it acts as pure QoS-weighted betweenness; the indented row restores the term ($\rho = 0.533$), so the defect slightly favors the baseline, and no comparative conclusion changes. Further baselines and per-fold values: Supplementary Section S43. Underlined: best predictor per column.

Ranker	LOSO ρ [95% CI]	Active $\rho_{>0}$	$\Delta\rho$ vs Topo-QoS [95% CI]	Folds won	p (p_{Holm})	Overlap@ K
<i>Reference: restatements of I^*’s propagation rule (not predictors)</i>						
Analytic I^*	0.808 [0.755, 0.853]	0.631	—	—	—	0.536
InDeg (direct dependents)	0.764 [0.674, 0.840]	0.516	—	—	—	0.504
Reach (transitive dependents)	0.732 [0.674, 0.782]	0.286	—	—	—	0.344
<i>Training-free baseline (registered comparator)$\P$</i>						
Topo-QoS	0.553 [0.443, 0.657]	0.280	—	—	—	0.388
corrected (articulation term restored)	0.533 [0.404, 0.650]	—	—	—	—	—
<i>Learned, on the dependency graph</i>						
GAT-P-QoS	<u>0.748</u> [0.704, 0.789]	<u>0.440</u>	+0.195 [+0.100, +0.294]	10/12	0.0068 $\ddagger$	<u>0.454</u>
HGT-P-QoS	0.514 [0.392, 0.636]	0.237	−0.039 [−0.155, +0.077]	4/12	0.622 $\ddagger$	0.380
<i>Learned and hybrid, on the raw multigraph</i>						
HGT-QoS	0.622 [0.547, 0.690]	0.312	+0.069 [−0.046, +0.174]	8/12	0.266	0.426
GAT-QoS	0.635 [0.567, 0.696]	0.338	+0.082 [−0.046, +0.201]	7/12	0.233	0.438
Hybrid-HGT	0.657 [0.572, 0.733]	0.345	+0.103 [+0.055, +0.152]	11/12	0.0034 (0.0068)	0.435
Hybrid-GAT	0.683 [0.603, 0.753]	0.362	+0.130 [+0.075, +0.190]	11/12	0.0015 (0.0029)	0.450

Hybrids beat the baseline, not their base learners. A hybrid takes the **Topo-QoS** score as an input and corrects its logit, so it nests its comparator. Hybrid-HGT (+0.103) and Hybrid-GAT (+0.130) beat **Topo-QoS** on 11 of 12 folds, also under the omnibus Holm correction ($p_{\text{omni}} = 0.041$ and 0.019), the Nadeau–Bengio corrected t ($p = 0.019$ and 0.014) and $|V_{\text{app}}|$ weighting. Neither differs from its own base learner (Hybrid-HGT vs. HGT-QoS +0.035, $p = 0.73$; Hybrid-GAT vs. GAT-QoS +0.048, $p = 0.30$), and the same holds with the articulation defect corrected (Table 7, F9). The hybrid gain therefore reflects the weakness of the comparator, not the learned correction. Given **InDeg** as their prior instead, both hybrids land within ± 0.012 of **InDeg** (F10): a learner handed the reference reproduces it.

Beyond the reachability oracle. On I_{dyn} , which agrees with I^* at $\rho = 0.711$, the references outperform all GNNs (first-order expansion 0.706, **InDeg** 0.664, GNNs 0.496–0.615; Table 6). Partial correlations show what each ranking keeps beyond I^* . After removing the rank of I^* , **InDeg** keeps 0.272 [0.188, 0.366], **Reach** 0.117, **Topo-QoS** 0.134 and the GNNs 0.107–0.173, all intervals excluding zero; after removing the first-order expansion instead, the raw-multigraph GNNs keep 0.212–0.217, some queue-flow signal beyond the unweighted first-order term but less than the rate-weighted reference (0.573).

What declared rates add on the queue-flow oracle. Weighting each term of the first-order expansion (0.706 on I_{dyn}) by the topic’s declared publication rate (Eq. (7)) raises it to 0.830 [0.778, 0.872] without training (exploratory). A gradient-boosted model trained on I_{dyn} labels, reading counts, the first-order expansion, declared rates and payload sizes, reaches 0.799: above the unweighted expansion (+0.094, Holm $p = 0.019$), but it falls below the rate-weighted reference, which exceeds it by +0.031 [+0.013, +0.049] on 10 of 12 folds (nominal Holm $p = 0.009$; exploratory, reporting the nominal p -value for this uncorrected contrast family). The difference is small, comparable to the per-fold variation that node order alone induces in the GNNs (Section 6.2), so we read it as the learned approximation not exceeding the formula rather than as a margin of superiority. In an exploratory attribution (Supplementary Section S41), the declared rate and payload columns alone add +0.097 (9/12 folds, Holm $p = 0.021$), and the seven QoS-derived columns ($w(t)$ -weighted scores and policy shares) add −0.005 ($p = 0.68$). Because I_{dyn} reads no payload (Section 4.3), that gain is rate signal; the payload columns cannot carry any. The GNN trained on the same labels, which received neither rates nor payloads, reaches only 0.598, below the same architecture trained on I^* and scored on I_{dyn} (0.615 as an ensemble). Because this GNN received topological features and QoS profiles without explicit rate inputs, this deficit reflects feature asymmetry as well as modeling capacity; it did not fit its target and is not a credible learned approximation of I_{dyn} , while a GNN given declared rates was not run.

Learning on top of the rate-weighted reference. The direct test of whether learning adds value beyond the aligned approximation is a learner that starts from it (Table 7, F12). Given Eq. (7) as one more input

Table 6: Rankers against three oracles, twelve LOSO folds, all 1,321 Applications (registered secondary and exploratory). Feature inputs differ: Eq. (7) and GBM read declared rates, whereas GNNs receive topology and QoS properties without rate features. Partial ρ : Spearman with I_{dyn} after the rank of I^* , or of the first-order expansion $\hat{I}_1^*$, is regressed out of both. The two learned blocks use different protocols: GNN rows trained on I^* score the mean of five seeds’ *predictions* (hence higher I^* values than Table 5’s per-seed means, e.g. 0.772 vs. 0.748), whereas rows trained on I_{dyn} (*, approximations of it) report the mean of per-seed ρ , so the two blocks are not directly comparable. Underlined: highest predictor per column among the rows shown; on I_{comp} , raw total degree (0.719, not tabulated) exceeds the underlined value.

Ranker	I^* ρ	I_{dyn} ρ [95% CI]	Partial $\rho(\cdot, I_{\text{dyn}} I^*)$ [95% CI]	Partial $\rho(\cdot, I_{\text{dyn}} \hat{I}_1^*)$	I_{comp} ρ
<i>Reference: restatements of I^*’s propagation rule (not predictors)</i>					
Analytic I^*	0.808	0.706 [0.629, 0.783]	0.318 [0.234, 0.412]	—	0.636
InDeg	0.764	0.664 [0.562, 0.758]	0.272 [0.188, 0.366]	0.079	0.650
Reach	0.732	0.583 [0.510, 0.651]	0.117 [0.055, 0.184]	0.102	0.302
<i>Reference: restatement of I_{dyn}’s first-order rule (not a predictor; exploratory)</i>					
Rate-weighted (Eq. (7))	0.756	0.830 [0.778, 0.872]	0.578 [0.471, 0.679]	0.573	0.551
<i>Training-free baseline</i>					
Topo-QoS	0.553	0.471 [0.381, 0.559]	0.134 [0.061, 0.204]	−0.091	<u>0.702</u>
<i>Learned GNNs (seed ensemble)</i>					
GAT-P-QoS	<u>0.772</u>	0.615 [0.549, 0.673]	0.111 [0.016, 0.213]	0.182	0.274
Hybrid-GAT	0.702	0.599 [0.514, 0.669]	<u>0.173</u> [0.082, 0.267]	0.056	0.585
Hybrid-HGT	0.672	0.573 [0.497, 0.638]	0.162 [0.074, 0.252]	0.029	0.582
HGT-QoS	0.667	0.549 [0.464, 0.622]	0.130 [0.035, 0.224]	0.212	0.201
GAT-QoS	0.647	0.523 [0.436, 0.606]	0.107 [0.018, 0.199]	<u>0.217</u>	0.144
HGT-P-QoS	0.618	0.496 [0.395, 0.601]	0.112 [0.033, 0.190]	0.087	0.334
<i>Learned approximations of I_{dyn} (trained on its labels)</i>					
GBM-P-QoS→dyn*	0.764	<u>0.799</u>	—	—	0.477
GAT-P-QoS→dyn*	0.734	0.598	—	—	0.368

column, the gradient-boosted approximation reaches 0.830, the formula’s value (+0.000, 5 of 12 folds, Holm $p = 0.97$); trained to correct the formula’s residual, it reaches 0.824 (−0.006, Holm $p = 0.68$); and the GAT trained on I_{dyn} with the formula as its prior gains +0.214 over the same GAT without it (12 of 12 folds) but still falls below the formula (0.812, −0.018, 1 of 12 folds, Holm $p = 0.0029$). The learned approximations recover the formula’s information and add none that is detectable here. On I_{dyn} this approaches equivalence: the interval of the gradient-boosted model given the formula ([−0.013, +0.015]) lies well inside ± 0.05 , unlike the comparison of GAT-P-QoS with InDeg on I^* .

How much is left to learn. The five-seed I_{dyn} label has an estimated reliability of $r = 0.79$ – 0.99 per fold, so no ranker can exceed $\sqrt{r} \approx 0.89$ – 0.996 (Section 4.3). The formula’s mean of 0.830 thus leaves roughly 0.06–0.17 of attainable agreement, depending on the fold. The learners started from the formula captured none of it. Whether that residual reflects mechanisms a learner could exploit with more data or is reachable only by simulating queueing is not resolved by this corpus (Section 7.5).

The multi-criteria oracle favors different rankers. On I_{comp} , learned rankers without a prior perform poorly (0.144–0.334), the hybrids keep 0.58, and training-free scores aligned with its fragmentation and flow terms rank highest (Topo-QoS 0.702, raw total degree 0.719), which again reflects overlap with the oracle’s terms. No ranker is best across all three oracles.

Identifying the critical set. Figure 5 shows the share of the true top-20% set that each ranker places in its top $k\%$, with ties resolved by averaging. At $k = 20\%$, GAT-P-QoS recovers 0.48 of the critical set and Topo-QoS 0.38; to recover 80%, GAT-P-QoS must flag the top 40% (0.81), and Topo-QoS does not reach 80% within 50%. The references do only slightly better: at $k = 20\%$, InDeg recovers 0.49 (tie-breaking bounds 0.48–0.50), and 80% recall requires the top 45% by InDeg (0.83) or by the first-order expansion (0.89). On I_{dyn} (panel B), GAT-P-QoS reaches 0.74 at 40% and 0.82 at 50%, compared to 0.80 at 40% for the InDeg reference; the rate-weighted reference, which restates I_{dyn} ’s first-order rule, recovers 0.81 of that set at 20% and 0.96 at 30% (exploratory).

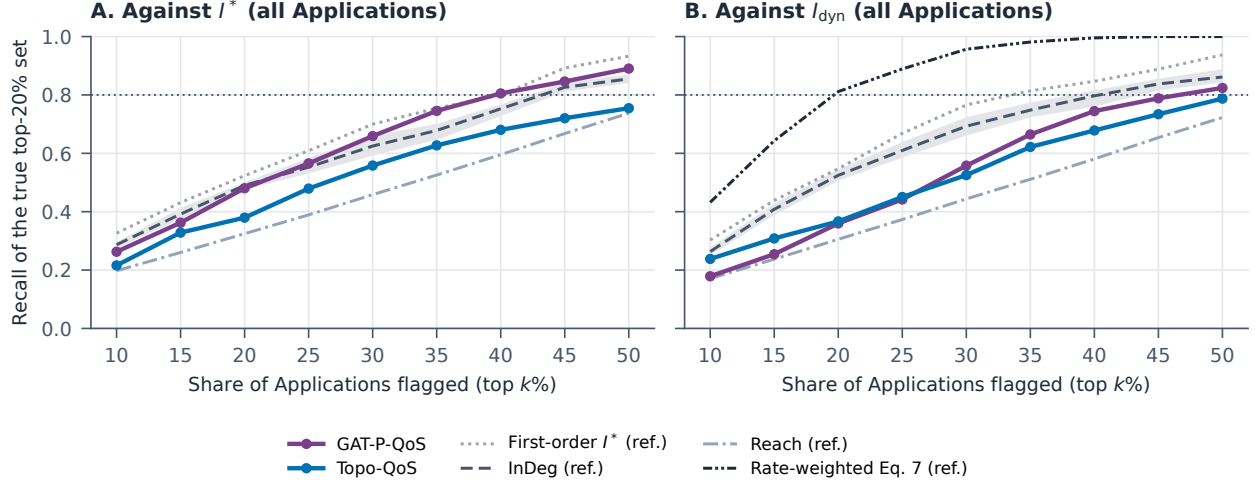


Figure 5: Recall of the true top-20% set (tie-inclusive) by each ranker’s top $k\%$, mean over the twelve LOSO folds, against I^* (A) and I_{dyn} (B), over all 1,321 Applications. Ties are resolved in expectation; grey dashed curves are references restating I^* ’s rule, the black dash-dotted curve in (B) is Eq. (7), and the shaded band gives InDeg’s tie-breaking bounds (Reach’s are much wider, 0.12–0.80 at $k = 20\%$). The dotted line marks 80% recall.

6.2. RQ2: Sources of Predictive Performance

Table 7 collects the registered controls; the full tables, with I_{dyn} , I_{comp} and zero-shot columns, are in Supplementary Section S42.

Table 7: Registered control arms (digest): twelve LOSO folds, five seeds, Application population, each run in one CPU invocation with every comparator re-run to an exact match. $\Delta\rho$ is paired by fold against the comparator, with a bootstrap 95% CI; p_{Holm} is within each family. $-\text{deg}$: `in_degree` and w_{in} zeroed. $-\text{R}$: every raw-graph edge also passed in reverse. $-\text{AP}$: articulation term restored. $-\text{min}$: every oracle-aligned feature zeroed (Section 3.5); $-\text{const}$: every node feature zeroed. F12 rows are trained and scored on I_{dyn} ; all others on I^* . F13 is a registered gate with a nominal p . Full tables: Supplementary Tables S69, S70 and S71.

Arm	Comparator	ρ	$\Delta\rho$ [95% CI]	Folds won	p_{Holm}
<i>F1, F2, F4: degree features, aggregator, and the 2×2 with w_{in} held</i>					
GAT-P-QoS–deg	GAT-P-QoS (0.748)	0.612	−0.136 [−0.205, −0.072]	1/12	0.0049
GIN-P-QoS–deg	GAT-P-QoS–deg	0.721	+0.108 [+0.027, +0.197]	8/12	0.157
“QoS” inputs (main effect)	averaged over T	—	+0.030 [−0.023, +0.085]	9/12	0.330
<i>F8–F10: edge direction, corrected prior, InDeg prior</i>					
GAT-QoS-R	GAT-QoS (0.635)	0.676	+0.041 [+0.000, +0.087]	10/12	0.064
GAT-P-QoS	GAT-QoS-R	0.748	+0.072 [+0.034, +0.113]	10/12	0.014
Hybrid-GAT-AP	Topo-QoS-AP (0.533)	0.669	+0.136 [+0.076, +0.198]	11/12	0.0059
Hybrid-GAT-AP	GAT-QoS	0.669	+0.034 [−0.047, +0.118]	7/12	0.940
GAT-QoS+InDeg	GAT-QoS	0.763	+0.128 [+0.063, +0.204]	11/12	0.0049
<i>F11: oracle-aligned features removed (last row descriptive)</i>					
GAT-P-QoS–min	GAT-QoS-R–min (0.378)	0.610	+0.231 [+0.128, +0.340]	10/12	0.0068
GIN-P-QoS–min	GAT-P-QoS–min	0.724	+0.115 [+0.038, +0.199]	9/12	0.027
GIN-P-QoS–const	InDeg (0.764)	0.719	−0.045 [−0.069, −0.024]	1/12	—
<i>F12: learners started from Eq. (7), scored on I_{dyn}; F13: node order permuted</i>					
GBM-P-QoS→dyn+Eq7	Eq. (7) (0.830)	0.830	+0.000 [−0.013, +0.015]	5/12	0.970
GAT-P-QoS→dyn+Eq7	Eq. (7)	0.812	−0.018 [−0.025, −0.012]	1/12	0.0029
GAT-P-QoS–perm	GAT-P-QoS	0.712	−0.035 [−0.055, −0.014]	3/12	(0.012)

The “QoS” factor was mostly the weighted in-degree. In the capacity-matched 2×2 on the raw multigraph (Supplementary Table S25), relation typing has no effect (−0.014, $p_{\text{Holm}} = 0.940$) and the “QoS” inputs add +0.073 ($p_{\text{Holm}} = 0.127$). The QoS-off arms, however, also zero w_{in} , a QoS-weighted version of InDeg; with w_{in} held in both arms (F4), the “QoS” effect drops to +0.030, so more than half of it was

the weighted in-degree. Closed-form controls agree: unweighted betweenness (0.591) and constant topic weights (0.595) match or exceed QoS-weighted betweenness (0.553), and permuting QoS profiles across topics has no effect ($\Delta = -0.006$, $p = 0.733$). Because messages never reach Applications on the raw multigraph (Section 3.4), this 2×2 compares per-node models and says nothing about relational typing (Section 7.2).

Degree features and the aggregator. Removing the in-degree and w_{in} columns lowers GAT-P-QoS by 0.136 (F1). This confirms known expressivity results rather than adding a new one: softmax attention averages over neighbors and cannot count them, whereas sum aggregation can [80, 81], and a GIN of the same size (GINE layers, which add the edge vector to each message) keeps 0.721 without the same columns (F2, not significant after Holm). In zero-shot transfer the degree columns matter little (0.810–0.821 without them), because the system models are ranked mostly by separating inert from active components.

Direction versus dependency semantics. The +0.113 gain of GAT-P-QoS over GAT-QoS could come from edge direction rather than dependency semantics, so a control passes every raw-graph edge in both directions, with shared weights and an unchanged parameter count (GAT-QoS-R, F8). It reaches 0.676: direction alone recovers +0.041 of the gain, which is not significant (Holm $p = 0.064$), and GAT-P-QoS still exceeds it by +0.072 [+0.034, +0.113] on 10 of 12 folds (Holm $p = 0.014$). The direction effect itself lies within the node-order spread reported below. Reverse edges also cost transfer (0.744 zero-shot, against 0.805–0.806; Table 8). This +0.072 representation gain illustrates the aggregator expressivity mechanism: on the raw multigraph, messages must traverse intermediate Broker and Topic nodes via 2-hop attention averaging (which normalizes neighbor weights and cannot count paths), whereas on the derived DEPENDS_ON projection, direct 1-hop sum aggregation or localized attention directly counts afferent dependents.

Without the oracle-aligned features. Every published learner reads features that compute part of I^* (Section 3.5). With all of them zeroed (F11), the raw-graph learners collapse (GAT-QoS to 0.369, the reverse-edge control to 0.378), whereas the dependency-graph GAT keeps 0.610 and beats the equally stripped reverse-edge control by +0.231 (10 of 12 folds, Holm $p = 0.0068$). The oracle-aligned set costs GAT-P-QoS 0.138, about what the two degree columns alone cost. Without them, sum aggregation beats attention on the dependency graph (+0.115, 9 of 12 folds, Holm $p = 0.027$), and without any node features at all, a sum-aggregation GNN on the dependency graph reaches 0.719, 0.045 below InDeg (1 of 12 folds): from structure alone it learns most of the dependent count, which the derived graph makes available and the raw multigraph does not. This too is what sum aggregation’s ability to count neighbors predicts [80, 82]; by precomputing the 2-hop composition into a direct 1-hop dependency link, the projection eliminates the need for the model to learn multi-hop neighbor counting across intermediate broker types.

Message passing, node order and the selection rule. On the raw multigraph, a gradient-boosted model on the same per-node features and no graph (GBM-Feat) reaches $\rho = 0.642$ under LOSO, level with GAT-QoS (0.635), and removing HGT’s reverse pass, its only route to Applications, has no effect (HGT-QoS-U, -0.010 , $p = 0.91$). ListMLE breaks label ties in input order (Section 4.2); permuting node order before training (F13; GAT-P-QoS-perm) scores 0.712 against 0.748 (-0.035 , 3 of 12 folds, $p = 0.012$), which triggered the registered flag. Two further permutations show variance rather than a favorable published order: seeds 17, 18 and 19 give 0.712, 0.740 and 0.734 (mean 0.729), and the per-fold spread across the three permutations averages 0.044. Because roughly 31% of components share tied zero labels ($I^* = 0$), arbitrary creation-order tie-breaking in ListMLE induces an empirical noise floor of approximately 0.044 per fold; differences between learned rankers within this envelope cannot be distinguished from permutation noise. Learned-ranker differences smaller than this spread should not be read as differences between models. Finally, the plan’s nested hyperparameter selection, applied later on the stage-1 grid, changes HGT-QoS from 0.622 to 0.677 (+0.055) and GAT-P-QoS from 0.748 to 0.693 (-0.054), neither significant (Holm $p = 0.259$); nested HGT-QoS versus Topo-QoS is +0.123 on 9 of 12 folds, still not significant (Holm $p = 0.192$). That harness trains on one scenario fewer and stops early on a held-out scenario, so it compares two protocols, not only two configurations (Section 5.3). Configuration swings of ± 0.055 are as large as several differences reported in this section, so every learned result here is conditional on one fixed configuration and on this data regime: eleven training architectures, about 1,000–1,300 labeled Applications per fold, and roughly 430,000 parameters per GNN. No learning curve over the number of training architectures and no matched tuning budget across model families was run (Section 7.5).

6.3. RQ3: Zero-Shot Transfer to Models of Open-Source Systems

Table 8: Zero-shot transfer to hand-authored models of five open-source systems (Application population), without fine-tuning; “zero-shot” applies only to the learned rows. All rows use the same labels. $\rho_{>0}$: rank correlation over active components. Reference rows restate I^* ’s rule. Intervals are percentile bootstraps over five systems, descriptive only. Underlined: best predictor per column.

Ranker	Evaluation Substrate	Mean ρ [95% CI]	Active $\rho_{>0}$	PR-AUC
<i>Reference: restatements of I^*’s propagation rule (not predictors)</i>				
Reach	Dependency graph	0.938 [0.879, 0.991]	0.871	0.933
InDeg	Dependency graph	0.863 [0.734, 0.952]	0.321	0.752
<i>Training-free baseline</i>				
Topo-QoS	Dependency graph	0.526 [0.357, 0.699]	−0.088	0.474
<i>Learned and hybrid, raw multigraph</i>				
HGT-QoS	Raw multigraph	0.760 [0.714, 0.819]	0.236	0.713
GAT-QoS	Raw multigraph	0.805 [0.759, 0.868]	0.319	0.790
Hybrid-HGT	Raw multigraph	0.695 [0.643, 0.730]	0.210	0.602
Hybrid-GAT	Raw multigraph	0.662 [0.597, 0.727]	0.185	0.600
<i>Learned, dependency graph</i>				
GAT-P-QoS	Dependency graph	<u>0.806</u> [0.785, 0.829]	<u>0.342</u>	<u>0.838</u>

On the five system models (Table 8), the best learned rankers reach about 0.81 (GAT-P-QoS 0.806, GAT-QoS 0.805), well above the training-free baseline (0.526), but the references rank higher still (Reach 0.938, InDeg 0.863). The difference is structural: half of the system models’ Applications have zero simulated impact (0.51, against 0.31 on the folds) and their fan-in is more concentrated (Gini 0.65 vs. 0.50), so separating the inert half already yields a high full-population ρ , and Reach separates it well. On the active stratum, every predictor is weak (learned 0.185–0.342, training-free baseline negative), as is InDeg (0.321), while Reach keeps 0.871: on these models, transitive reach is nearly what I^* computes. The baseline prior reduces transfer: both hybrids fall below their base learners (0.695 and 0.662 against 0.760 and 0.805), also with the prior’s defect corrected (0.702 and 0.668). Because all five models were written by one modeler (Section 5.1), these results show that learned rankers transfer broad topological partitioning across one modeler’s style, not reliable discrimination among active components in new architectures. Consequently, neither learned rankers nor analytical counts should be deployed as fine-grained prioritizers within the active component stratum without operational failure logs or runtime telemetry.

6.4. RQ4: Cost

Table 9: Like-for-like cost, every stage timed on the same graph in one session (CPU, one thread, median of 3 runs; once at 4,995 components, where the three-type sweep was not run). Count: Application–Library projection plus InDeg. One I^* pass: seed 42, Applications. Sweep: the published five-seed labeling run over Applications, Brokers and Libraries. Features: the analysis producing the learned rankers’ node features. Gate: system-layer analysis with 18 anti-pattern detectors, for reference. Corpus row: minimum–median–maximum over the twelve folds.

$ V $	Count (ms)	One I^* pass (s)	Sweep (s)	Features (s)	Gate (s)	Pass : count	Features : pass
249	2.5	0.26	1.63	1.46	2.05	105×	5.7×
499	6.6	0.79	6.59	6.49	10.01	119×	8.2×
999	13.1	3.90	32.42	30.57	50.92	297×	7.8×
1,998	35.5	18.45	157.86	160.73	275.80	521×	8.7×
4,995	226.2	94.58	—	1,768.81	3,072.60	418×	18.7×
Corpus	0.4–15.6	0.01–0.72	0.09–4.98	0.07–52.25	0.16–81.48	17–45–176×	4.5–16.9–72.5×

Measured like for like (Table 9), one labeling pass of the reachability oracle over a corpus architecture’s Applications takes 0.01–0.72 s, the counting path that restates its first wave is 17–176× cheaper (median

45 $\times$), and the feature extraction every learned ranker needs 4.5–72.5 $\times$ more (median 16.9 $\times$); the ordering holds on generated graphs of up to 5,000 components. The articulation and CDI phase accounts for 88–91% of feature extraction, so this cost belongs to the chosen feature set, not to learned ranking as such: a sum-aggregation GNN with no node features reaches 0.719 (Table 7, F11) and would not pay it. Where the I^* ranking itself is wanted, running the oracle is therefore cheaper than approximating it with a learned ranker at every size measured. Inference is negligible (tens of milliseconds at 2,000 components); the count scales to 10,000 components (1.1 s), **Reach** less well (23 s).

The computationally intensive oracle is I_{dyn} : labeling the corpus took 12.7 CPU-hours (about 7 s per fault run and seed). Crucially, this execution cost stems from SimPy’s discrete-event simulation interpreter overhead rather than intrinsically complex physical dynamics; the rate-weighted reference captures the underlying first-order delivered-rate loss in at most about a millisecond per architecture. Estimating energy as wall-clock time multiplied by the processor’s 28 W base power (Section 7.4), labeling the corpus with I_{dyn} took about 355.6 Wh, the five-seed I^* labeling sweep 11.1 s (0.086 Wh), and the full detection gate 106.9 s (0.83 Wh). Summed over the twelve folds, the feature extraction that the learned rankers read took 70.0 s (about 0.54 Wh), and the dependency count 0.037 s (about 1 J). Training is a separate one-off cost per model version: the last sweep with recorded wall-clock took 7.7 CPU-hours for four learned arms of 60 fits each (Supplementary Section S32), about 0.22 kWh by the same estimate. At these magnitudes, raw per-inference energy differences are practically negligible; what governs operational feasibility in continuous integration (CI/CD) is the amortized lifecycle break-even. A learned model incurs an upfront cost for simulator label generation (about 30 Wh per corpus architecture, 355.6 Wh over twelve) plus training (0.22 kWh), which must be amortized against recurrent CI evaluation runs. Because the training-free rate-weighted reference incurs neither labeling nor training overhead while achieving equal or superior ranking accuracy, it dominates learned approximations across any CI/CD pipeline lifecycle.

7. Discussion, Threats to Validity, and Limitations

7.1. Dependency Representation versus Model Complexity

For the reachability and queue-flow simulators in this corpus, explicit dependency representations accounted for more of the ranking performance than model complexity did. The derived dependency graph enabled both analytical and learned approaches: attention-based learners improved on it relative to the raw multigraph, and the improvement survives both a control for edge direction and the removal of every feature that computes part of the reachability simulator. Yet rankings computed directly from the same graph were not significantly different from, or exceeded, the best learned models. In this corpus, explicit dependency derivation appears to be the principal source of predictive signal, not a preprocessing step for learning. The multi-criteria simulator is the exception: there, raw degree and betweenness, whose terms align with fragmentation, ranked highest.

Several of the highest-scoring rankings largely restate a simulator’s propagation rule. The reference criterion (Section 4.4) separates recovering a simulator’s definition from predicting its outcomes, and it changes how learned results should be read: a learned model that approaches a reference has learned the simulator’s first-order rule, not necessarily anything beyond it. Because low-order simulators admit low-order truncations by definition, this study’s negative findings are bounded strictly to first-order simulation regimes where publish–subscribe afferent coupling dominates failure propagation. Importantly, this negative result should not be misconstrued as an assertion that graph neural networks lack predictive utility across all distributed systems dependability tasks. In regimes characterized by non-first-order dynamics—such as cyclic backpressure loops, cascading retry storms, buffer bloat, or Byzantine failures—multi-hop message-passing GNNs possess expressive capacity that simple local degree counts cannot reproduce; whether GNNs outperform analytical surrogates in such non-first-order regimes remains an open empirical question. Because simulation-labeled benchmarks are common in architecture analysis, the criterion offers a general check: before crediting a learned model with predictive skill, compare it with the simplest ranking that restates the simulator’s own computation.

7.2. When Learning Adds Value, and Relative to What

Relative to the training-free baseline, learning helped in two ways, both weak evidence for learning. Hybrids outperformed the baseline they nest but not their own base learners, so the gain belongs to the baseline’s weakness; and learned models transferred zero-shot better than the baseline, mainly by separating inert from active components. Relative to analytical references built on the same dependencies, no learned model showed additional skill, and the most direct test, learners that start from the rate-weighted approximation, found no improvement on it. The learned queue-flow approximation beat the unweighted expansion only because it read declared rates; once rates enter the analytical expansion, the deficit is not one of missing inputs. Learning is likely to contribute most where cascade impact depends on multi-hop, non-linear mechanisms that no low-order approximation captures, such as broker queue congestion, backpressure onto publishers, retry storms or heterogeneous service capacities. None of the three simulators implements them (Section 4.3). Under the first-order reachability and rate-loss mechanisms they do implement, explicit dependency derivation supplied the structural signal directly, and learners trained on eleven architectures in one fixed configuration added none beyond it. Whether learning helps where those mechanisms are present, or with many more training architectures, is untested here (Section 7.5). A per-regime breakdown by architecture type is given in Supplementary Section S36.

How message passing contributes. On the raw multigraph, learning used node properties, not relations: a graph-free model on the same features matches the GAT, and removing HGT’s only route to Applications has no effect (Section 6.2). On the derived dependency graph, message passing gathers the dependent neighborhood; attention still needs the degree columns to approach the dependent count, whereas sum aggregation recovers most of it from structure alone. These observations confirm known results on the expressivity of aggregators [80, 81, 82]; their contribution here is to show that they decide whether a learner can reach the dependency-count reference.

What the study can say about relation typing. Little. On the raw multigraph, where typing was compared at matched capacity, messages never reach Applications, so that comparison is between per-node models. On the dependency graph the projection has only two relation types, and the heterogeneous transformer (HGT-P-QoS) ran in one untuned configuration in which some seeds converged to rankings anti-correlated with the oracle (mean within-fold seed standard deviation 0.254, mean $\rho = 0.514$), while GAT-P-QoS was stable (0.030, $\rho = 0.748$). That is an optimization failure of one configuration, not evidence about typing; testing typing would need a substrate with active message passing, more relation types and a tuned heterogeneous model.

7.3. Practical Consequences

Choosing a ranker. Table 10 summarizes the evidence by failure notion; it is conditional on simulation fidelity, and neither simulator, nor any approximation of it, has been shown to predict real outages. If the simulator is cheap, like I^* , its ranking can be computed directly from the manifest, and publish–subscribe afferent coupling [64, 65] costs a fraction of that; its value is cost, not predictive skill. If the simulator is expensive, like I_{dyn} , the rate-weighted expansion approximates it in milliseconds, and declared publication rates, not QoS policies, are the only declared information that carried signal. Training-free dependency analysis is therefore also the lower-energy choice at commit and staging time [22, 21, 20].

Closed-form rankings are also deterministic, whereas learned rankings vary with the seed (within-fold standard deviation 0.024–0.030 for the GATs, up to 0.254 for the unstable HGT-P-QoS) and with node order (Section 6.2). Learned models transfer zero-shot better than the training-free baseline mainly by separating inert from active components, so they should not be relied on to order the components that actually propagate failures ($\rho_{>0} \leq 0.342$). Practitioners should treat static structural rankers strictly as coarse-grained filters for isolating inert components, and avoid deploying them as fine-grained prioritizers within active components in unfamiliar systems unless supplemented by operational traces or failure histories.

A two-tiered triage protocol. Recovering 80% of the true critical set takes the top 40–45% of Applications by the best rankers, and no predictor reaches 90% within the top 50% (Figure 5). A gate that catches most critical components would flag nearly half the system, so these rankers are unsuitable as automated pass/fail filters. We instead propose a two-tiered triage protocol as an inspection heuristic, which needs human-factors and workflow validation before adoption:

Table 10: Which ranker the evidence supports, by the failure notion the oracle encodes (registered secondary and exploratory; conditional on simulation fidelity). Where a reference restating the simulator’s rule exists (Section 4.4), it is recommended because it is cheaper and at least as accurate in this corpus, a descriptive comparison; it is not credited with predictive skill.

Failure notion (oracle)	Ranker	Evidence
Reachability cascade (I^*)	If I^* is accepted as the impact definition, run I^* directly	0.01–0.72 s per pass; best predictor GAT-P-QoS $\rho = 0.748$, not significantly different from the direct-dependent count (0.764); equivalence not established
Queue-flow delivery loss (I_{dyn})	Rate-weighted expansion (training-free; exploratory)	$\rho = 0.830$ in milliseconds (Eq. (7); exploratory); gradient-boosted approximation 0.799, and learners started from the formula do not exceed it; unweighted expansion 0.706
Fragmentation and throughput (I_{comp})	Raw total degree (training-free)	$\rho = 0.719$, term-aligned with fragmentation; Topo-QoS 0.702 (its articulation term is zero by a defect); among learned models, only hybrids retain signal ($\rho = 0.585$), while pure learners collapse (≤ 0.334)
Unfamiliar topology	GAT-P-QoS or GAT-QoS	$\rho = 0.806 / 0.805$ vs. baseline 0.526; dependency counts rank higher (0.863–0.938); filters inert nodes, but active discrimination is weak ($\rho_{>0} \leq 0.342$); five single-author models

1. **Tier 1 (commit stage):** compute the direct-dependent count and the rate-weighted expansion (milliseconds; the latter exploratory) and show the top five components by each as an informational note in pull-request review, without blocking developers.
2. **Tier 2 (staging and sprint review):** run the simulators on the Tier-1 shortlist within a fixed inspection budget, such as the ten highest-ranked services, and reserve redundant broker routing or chaos fault injection for the components the simulation confirms.

7.4. Threats to Validity

Construct validity and oracle circularity. All labels are simulated: no ranker is validated against observed failures, so the findings establish what can be predicted within the evaluated simulators, not what predicts real failures. The simulators omit runtime dynamics such as buffer bloat, thread-pool exhaustion, garbage-collection pauses and retry storms, and some real outages are sustained by load feedback rather than structure [48, 49]. The circularity is analyzed in Section 4.4: because the simulation oracles are low-order by construction, low-order analytical truncations under Simplifications S1–S5 capture their primary signal (I_{dyn} propagates no failure beyond one hop; Section 4.3). I^* and I_{dyn} are well approximated by low-order functions of the dependency graph (a first-order expansion recovers 0.808 of I^* and a rate-weighted one 0.830 of I_{dyn}), and I_{comp} ’s terms track degree and betweenness, which rank highest on it (0.70–0.72). The benchmark therefore did not exhibit a regime in which learning exceeds an aligned analytical ranking. Its absence here is a property of these oracles, not evidence that none exists, and the study’s negative answer is bounded accordingly. The reverse-edge control shares one set of weights across both directions; a direction-typed variant was not tested. Rules 2–4 and 6 are untested.

Internal validity and reproducibility. No ranker reads simulator output (regression tests verify this). Capacity, depth and early stopping were held constant across arms, but every learned result is conditional on one fixed configuration (Section 5.3); the registered nested selection, run later, moved learned rankers by up to ± 0.055 . The learners were also trained in a small-data regime (eleven architectures, about 1,000–1,300 labeled Applications per fold, roughly 430,000 parameters), which biases a learned-versus-analytical comparison toward the null; sample complexity across larger synthetic corpora was not evaluated. The learned rankers are seed-sensitive: the mean within-fold standard deviation of ρ is 0.099 for **HGT-QoS**, 0.254 for **HGT-P-QoS** and 0.024–0.030 for the **GATs**, and one of 60 **HGT-QoS** runs and four of 60 **HGT-P-QoS** runs converge to rankings anti-correlated with the oracle; they are kept in the per-seed means, as registered. Due to non-deterministic scatter/gather operations in PyTorch Geometric across different CPU/GPU hardware architectures and linear algebra libraries, an unconstrained learned cell exhibited numerical drift of up to 0.172 across distinct

environments. Every reported learned-versus-learned contrast is therefore strictly paired within self-contained sweeps where all comparators are re-run simultaneously under identical hardware; paired contrasts were not repeated across different devices. Node order adds a variance of about 0.04 per fold (Section 6.2). CDI uses hash-dependent tie-breaking, so `PYTHONHASHSEED=0` is fixed, and all reported values are reconciled mechanically against released artifacts.

External validity and the single-modeler threat. Synthetic topologies come from a single generator family, which may induce regularities not representative of industrial architectures. The five system models (22–41 Applications) were each written by the first author from public documentation; no second modeler authored any of them, and no automated manifest extractor was used. They also differ structurally from the synthetic folds in ways that favor **Reach** (Section 6.3), and two are RPC systems re-expressed as publish–subscribe meshes. Moreover, these models feature a bimodal distribution with over 50% inert components ($I^* = 0$), so strong full-population ρ reflects separating inert nodes rather than fine-grained discrimination among active spreaders ($\rho_{>0} \leq 0.342$). No topology was extracted from a real deployment manifest: the released tooling has no importer for launch files or for compose or Kubernetes manifests, and the five models are encoded by hand. No fault-injection experiment on a deployed system was run, and the replication package’s re-modeling protocol, which reports per-type Jaccard agreement between independently authored models (`reproduce/model_agreement.py`), has not yet been executed by an independent modeler. We timed the counting path up to 10,000 components but did not evaluate rankers or oracles at that scale. We did not reproduce published learned criticality models (FINDER, DrBC), which do not support typed multigraphs or pub-sub semantics.

Energy estimation and Green AI. Energy figures are wall-clock time multiplied by the processor’s 28 W base power (Intel Core i7-1370P; `reproduce/energy_estimate.py`), because RAPL counters required privileges unavailable to us. This is a nameplate estimate, not a bound in either direction: single-threaded work may draw less than the package’s base power and turbo frequencies more, and DRAM and idle power are ignored. At the magnitudes involved, fractions of a watt-hour per corpus for every training-free and learned ranking against 355.6 Wh to label the corpus with I_{dyn} , the energy argument of this paper rests on the break-even of a learned I_{dyn} approximation (Section 6.4), not on differences between rankers [22, 21]. The feature-extraction cost of the learned rankers belongs to the chosen feature set, not to learning as such.

Conclusion validity. Only the plan’s two co-primary contrasts are confirmatory, and both were null (Section 5.3). LOSO folds share ten of eleven training scenarios, so Wilcoxon p -values are anti-conservative; the Nadeau–Bengio corrected test leaves the hybrid contrasts significant and the primary null. Twelve folds from one generator make the study underpowered for equivalence at ± 0.05 : it is not established for any pair of independently derived rankers, and holds only for learners given a reference as their prior, which reproduce it (Table 7, F10); on I_{dyn} , the interval of the learner started from Eq. (7) also lies within that margin. Where the paper says one ranker “matches” another, it means the two are not significantly different (Section 5.3).

7.5. Limitations and Future Work

Five directions follow, in order of priority:

1. **A confirmation corpus:** scenarios generated after the analysis is frozen, on which the reference criterion and the registered secondary and exploratory claims are re-tested. Because each topology regenerates exactly from its configuration and labeling with I^* takes seconds, this is the cheapest way to make the present headline findings confirmatory.
2. **Real topologies, real failures and independent modelers:** automatic extraction of SaG models from real artifacts (ROS 2 launch files with HAROS or ROSDiscover [62, 63], docker-compose and Kubernetes manifests); controlled fault injection on ROS 2, Kafka and MQTT deployments and on reference systems with known faults such as Train-Ticket [32]; distributed tracing to check whether declared dependency paths match runtime message flow; and re-authoring of the five system models by additional modelers with agreement reported.
3. **Non-first-order impact models:** queue-flow variants with backpressure onto publishers, finite broker capacity, retry amplification and heterogeneous service capacity, and simulators with synchronous call semantics, where an aligned analytical approximation may not exist. These are the regimes in which

learning could exceed analytical methods, and the only ones in which this study’s title question could receive a positive answer. In such regimes, higher-order multi-hop GNN message passing may prove essential to capture complex cyclic or non-linear failure dynamics that defeat low-order topological truncations.

4. **The learning regime:** learning curves over the number of training architectures (the generator makes these free), a matched tuning budget for every model family, smaller models, a tie-aware listwise loss, and a GNN given declared rates for the queue-flow oracle.

5. **Brokers and hosts:** evaluating Rules 2–4 and 6 by ranking Brokers and Hosts.

Further extensions include measured energy (RAPL or IPMI) in place of the nameplate estimate, a queue-flow run with QoS policies disabled, a payload-aware queue-flow oracle (registered but not run), the sensitivity of I_{comp} to its declared weights, and validation of the proposed explanation layer.

8. Conclusion

This paper presented Software-as-a-Graph (SaG), a pre-deployment framework that models explicit dependency graphs from publish–subscribe architecture specifications and deployment descriptors, supporting analytical, hybrid, and learned approaches to cascade-impact ranking. Using twelve synthetic architectures, three failure simulators, and five stylized open-source system models, the study examined when graph learning improves rankings derived directly from explicit architectural dependencies. The registered co-primary contrasts were null, and the findings below are registered secondary or exploratory.

For the reachability and queue-flow simulators, which are dominated by first-order propagation by construction, explicit dependency representations provided most of the predictive signal observed in the evaluated corpus. Attention-based learners benefited from the derived dependency graph ($\rho = 0.748$ vs. 0.635 on the raw multigraph), a gain that survives controls for edge direction and for features that compute part of the simulator, but simple analytical rankings built on the same dependencies did not differ significantly from, or exceeded, the learned models. Counting direct dependents ($\rho = 0.764$) was not significantly different from the best learned model on the reachability simulator, and on the expensive queue-flow simulator a rate-weighted first-order approximation computed in milliseconds (0.830) exceeded a learned approximation trained on 12.7 CPU-hours of simulation labels (0.799; exploratory); learners that start from that approximation matched it at best. In this simulated corpus, representation mattered more than model complexity.

Learning outperformed the training-free baseline but not the strongest analytical references. Hybrids that correct a closed-form prior significantly outperformed that baseline, but not their own base learners, and learned models transferred zero-shot to unfamiliar architectures better than the baseline (≈ 0.81 vs. 0.53). In both cases, dependency-based references remained competitive or stronger, and ordering among components that actually propagate failures remained weak. These conclusions hold for learners trained on eleven synthetic architectures in one fixed configuration, against simulators rather than observed failures.

These results have implications for both research and practice. For researchers, they show the importance of evaluating graph-learning approaches against analytical alternatives aligned with the simulator, and of distinguishing prediction from restatement of a simulator’s rule. For practitioners, they suggest that training-free dependency analysis provides effective and inexpensive support for prioritizing architectural review. Learned models can be expected to add value only where no analytical approximation aligned with the simulator is available. Each simulator studied here is closely tracked by such an approximation, so this condition did not arise in this corpus.

Sections 7.4 and 7.5 discuss limitations, including the reliance on simulated failures; re-testing these findings on a corpus generated after the analysis was frozen, under non-first-order simulators and against real outages, is the natural next step. More broadly, the study suggests that the central challenge in architecture-level cascade-impact analysis is not selecting increasingly sophisticated learning algorithms, but deriving explicit dependency representations and understanding when learning adds value beyond them. By making this comparison explicit and releasing a reproducible benchmark, Software-as-a-Graph provides a foundation for more principled evaluation of analytical and learned methods for pre-deployment architecture analysis.

Declarations

CRedit Authorship Contribution Statement. Ibrahim Onuralp Yigit: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. Feza Buzluca: Conceptualization, Methodology, Validation, Writing – review and editing, Supervision.

Declaration of Competing Interest. The authors declare no competing financial interests or personal relationships that could have influenced this work.

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Availability. The complete replication package—including datasets, simulation harnesses, model checkpoints, analysis scripts, and reproduction workflows—is archived on Zenodo under DOI 10.5281/zenodo.23045204 [102]. The software [103], environment specifications, and reproduction documentation are publicly available on GitHub at an immutable commit, <https://github.com/onuralpyigit/software-as-a-graph/tree/56d9bff8ae9583ee9df1d270f0650a3a7c3239e8/docs/research/jss/experiments>. The script `reproduce/reconcile_manuscript.py` mechanically verifies 1,817 reported figures against the released artifacts.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work, the authors used Anthropic’s Claude Opus 5 and Opus 5.5, and Google’s Gemini Flash 3.8, to suggest revisions to the manuscript text, which the authors evaluated, edited, and verified against the results; for LaTeX typesetting and formatting assistance; and for developing analysis scripts in the replication package. They used Grammarly for grammar checking and language copy-editing. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article. The study design, the choice of experiments, the interpretation of results, and all scientific claims are the authors’ own.

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